

HCIR for Swiss Open Government Data Using Fuzzy Logic-Based Ranking

Master's Thesis

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Abstract

Open Government Data (OGD) portals provide access to large collections of publicly available datasets that support transparency, innovation, and evidence-based decision-making. However, users frequently encounter difficulties locating relevant datasets because conventional keyword-based retrieval methods primarily rely on lexical similarity and often fail to exploit the rich metadata associated with published datasets.

This thesis proposes a Human-Centred Information Retrieval framework for the Swiss Open Government Data portal that integrates lexical relevance with metadata completeness, dataset recency, and resource availability using a Mamdani fuzzy inference system. The framework combines multiple metadata-derived relevance criteria within an interpretable ranking model and generates human-readable explanations for ranking decisions while maintaining competitive retrieval effectiveness [42, 24, 4, 21].

The proposed framework was implemented as a prototype and evaluated using a reproducible retrieval benchmark consisting of a frozen metadata snapshot, fifteen representative search queries, and standard information retrieval metrics, including Mean Average Precision (MAP), Precision@5, Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain (nDCG@10) [25, 16]. The evaluation compared the proposed fuzzy ranking approach with the existing portal ranking, a BM25 lexical baseline, a weighted linear metadata baseline, and a semantic retrieval baseline. A complementary formative user evaluation involving ten Master’s students assessed the perceived clarity, transparency, and usefulness of the explanation component.

The results demonstrate that the proposed framework achieved competitive retrieval effectiveness while providing interpretable, metadata-driven ranking explanations. No statistically significant pairwise superiority was observed after Holm–Bonferroni correction, indicating comparable performance across the evaluated retrieval systems. The findings suggest that fuzzy logic provides a practical mechanism for supporting explainable metadata-aware retrieval within Human-Centred Information Retrieval without sacrificing competitive retrieval performance.

This research contributes a transparent HCIR framework, a reproducible evaluation methodology, and a prototype demonstrating how fuzzy logic and metadata-driven explanations support exploratory dataset discovery in Swiss Open Government Data portals. The proposed approach provides a foundation for future research on explainable, metadata-aware retrieval systems for public-sector information discovery.

Keywords: Human-Centred Information Retrieval, Open Government Data, Fuzzy Logic, Mamdani Fuzzy Inference System, Metadata-Aware Ranking, Explainable Information Retrieval, Information Retrieval Evaluation, Swiss Open Government Data.

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List of Abbreviations

API	Application Programming Interface
BM25	Best Matching 25
CKAN	Comprehensive Knowledge Archive Network
DCAT	Data Catalog Vocabulary
DSR	Design Science Research
HCIR	Human-Centered Information Retrieval
HCI	Human-Computer Interaction
IR	Information Retrieval
NLP	Natural Language Processing
OGD	Open Government Data
SBERT	Sentence-BERT
TF-IDF	Term Frequency-Inverse Document Frequency
UI	User Interface
XAI	Explainable Artificial Intelligence
XIR	Explainable Information Retrieval

Chapter 1

Introduction

The rapid growth of OGD has increased opportunities for transparency, innovation, and evidence-based decision-making. Governments worldwide have made large volumes of public-sector data openly accessible through national data portals, enabling researchers, businesses, policymakers, and citizens to access and reuse information across domains such as transportation, health, education, environment, and public administration. As these repositories continue to expand, however, discovering relevant datasets has become increasingly difficult. Traditional keyword-based retrieval systems often struggle to satisfy complex or exploratory information needs, highlighting the need for retrieval approaches that better support human search behaviour.

This chapter introduces the research problem addressed in this thesis, identifies the research gap, presents the objectives and research questions, summarizes the main contributions, and outlines the structure of the thesis.

1.1 Background

OGD initiatives have become an important component of digital government strategies worldwide. By making public-sector data openly available, governments aim to improve transparency, accountability, citizen participation, and innovation [15, 1]. Open data enables researchers, businesses, policymakers, and citizens to access and reuse governmental information for scientific research, policy analysis, economic development, and public service innovation.

Many countries have established national open data portals that aggregate datasets published by public institutions. These portals provide centralized access to datasets originating from multiple governmental organizations and domains, including transportation, health, education, environment, energy, public administration, and demographics [1]. The availability of such datasets creates significant opportunities for evidence-based decision-making and data-driven innovation.

Switzerland actively supports OGD initiatives through the national portal *open-data.swiss*, which serves as a centralized platform for accessing datasets published by federal, cantonal, and municipal organizations [10]. The portal currently hosts thousands of datasets covering a wide range of thematic domains and provides access through both web interfaces and application programming interfaces (APIs).

Despite the increasing availability of public datasets, effective dataset discovery remains a significant challenge. As the number of available datasets grows, users often struggle to identify datasets that satisfy their information needs. The challenge is not only related to the quantity of available data but also to the complexity of metadata descriptions, inconsistencies across publishers, varying levels of metadata quality, and differences in terminology used to describe similar concepts [1].

Traditional dataset retrieval systems primarily rely on keyword-based search mechanisms. These systems typically rank datasets based on textual similarity between user queries and dataset metadata. Although such approaches are computationally efficient and widely adopted, they often fail to capture the complexity of human information needs. Users frequently express vague, exploratory, or partially defined queries that cannot be effectively interpreted using strict keyword matching alone [26].

This challenge becomes particularly important within the field of HCIR, which emphasizes the interaction between users and retrieval systems. Rather than focusing solely on algorithmic ranking performance, HCIR seeks to design retrieval systems that align with human reasoning, support exploratory search behavior, and provide understandable explanations for retrieval decisions [26].

In this context, fuzzy logic presents a promising approach for modeling uncertainty and gradual relevance. Unlike traditional binary retrieval models that classify documents as either relevant or non-relevant, fuzzy logic enables relevance to be represented as a continuum. This capability makes fuzzy systems particularly suitable for handling vague concepts and supporting human-centered retrieval scenarios where relevance is inherently subjective and context-dependent [42, 21].

1.2 Problem Statement

The rapid growth of OGD repositories has significantly increased the complexity of dataset discovery. Although large quantities of data are available through platforms such as *opendata.swiss*, users frequently encounter difficulties when searching for relevant datasets.

Existing retrieval approaches primarily rely on lexical relevance and provide limited support for transparent multi-criteria ranking and exploratory search. These approaches estimate relevance based on statistical relationships between query terms and document metadata [35, 34]. While effective for exact-match retrieval scenarios, they often exhibit

several limitations when applied to exploratory dataset search.

First, traditional retrieval models are highly sensitive to vocabulary mismatch. Users may describe an information need using terminology that differs from the terminology used within dataset metadata. As a result, potentially relevant datasets may not appear among the top-ranked results.

Second, conventional ranking systems typically focus on textual similarity while neglecting other important metadata characteristics. Factors such as dataset recency, metadata completeness, update frequency, and thematic relevance may significantly influence dataset usefulness but are often not incorporated directly into ranking decisions.

Third, many retrieval systems operate as black-box mechanisms that provide little transparency regarding how relevance scores are computed. Users may receive ranked results without understanding why certain datasets were recommended or why others were excluded. This lack of transparency can reduce trust in retrieval systems and hinder effective decision-making [8].

Finally, human information needs are often vague, incomplete, and evolving. Users exploring OGD frequently begin with broad topics rather than precise dataset requirements. Traditional retrieval systems are not well equipped to support such exploratory search behavior because they assume clearly defined queries and exact matching conditions [26].

These limitations indicate the need for retrieval approaches that can integrate multiple relevance criteria, accommodate uncertainty, and provide transparent explanations for ranking decisions.

1.3 Research Gap

The literature identifies three dominant categories of retrieval approaches: keyword-based retrieval, semantic retrieval, and fuzzy retrieval.

Keyword-based retrieval approaches, including TF-IDF and BM25, remain widely used due to their efficiency and simplicity. However, they primarily rely on lexical matching and may struggle to address ambiguity, vocabulary mismatch, and exploratory search behavior [35, 34].

Recent advances in Natural Language Processing (NLP) have introduced semantic retrieval methods based on embedding models such as Sentence-BERT (SBERT). These approaches improve the ability to capture semantic relationships between queries and documents by representing text within continuous vector spaces [33]. While semantic retrieval reduces vocabulary mismatch, it often introduces additional complexity and reduced interpretability.

In contrast, fuzzy Information Retrieval (IR) provides mechanisms for modeling gradual relevance and uncertainty. Through fuzzy membership functions and rule-based reasoning, retrieval systems can represent relevance as a degree rather than a binary condi-

tion. Furthermore, fuzzy inference systems provide explicit reasoning structures that can be translated into understandable explanations [21, 42].

Although previous studies have investigated keyword-based retrieval, semantic retrieval, and fuzzy IR independently, the fundamental challenge remains insufficiently addressed. Existing approaches generally optimize lexical or semantic matching but provide limited support for combining multiple metadata-based relevance criteria within an interpretable decision-making process. As a result, users often receive ranked results without a transparent explanation of why particular datasets are prioritised, which is especially problematic in public-sector information systems where accountability and trust are important.

The research gap addressed in this thesis is therefore not merely the absence of a particular combination of techniques, but the lack of a human-centered retrieval framework that integrates multiple metadata-derived relevance signals while preserving transparency and explainability. This thesis addresses that gap through the design, implementation, and evaluation of a fuzzy logic-based HCIR framework for Swiss OGD, enabling interpretable multi-criteria ranking that supports exploratory dataset discovery and provides understandable explanations for ranking decisions.

1.4 Research Objectives

The primary objective of this thesis is to design, implement, and evaluate a fuzzy logic-based retrieval framework that improves dataset discoverability within the Swiss OGD portal.

The specific objectives are:

1. To analyze limitations of existing OGD retrieval approaches.
2. To design a metadata-driven retrieval framework capable of integrating multiple relevance criteria.
3. To develop a fuzzy inference model that supports gradual relevance estimation.
4. To generate interpretable explanations for ranking decisions.
5. To evaluate the effectiveness, usability, and transparency of the proposed system through quantitative and qualitative methods.
6. To investigate the contribution of fuzzy logic to HCIR.

Beyond defining the technical objectives of the proposed framework, it is equally important to identify the intended users and explain how their requirements influenced the overall system design. The proposed framework is primarily intended to support

researchers, public-sector analysts, journalists, students, and citizens seeking to discover and understand Swiss OGD. These user groups frequently perform exploratory searches in which information needs may be incomplete, ambiguous, or evolve during the search process. Consequently, the framework was designed according to HCIR principles, emphasizing transparent ranking decisions, explainable retrieval results, and the integration of multiple relevance criteria beyond keyword matching. These design choices aim to improve users' understanding of why datasets are retrieved and support more informed exploration of public-sector information.

1.5 Research Questions

The study is guided by the following research questions:

RQ1: How can fuzzy logic be used to improve dataset retrieval within OGD portals?

RQ2: How can multiple metadata-based relevance criteria be integrated into a unified and interpretable ranking mechanism?

RQ3: To what extent does a fuzzy logic-based retrieval system support HCIR principles?

RQ4: How effective are metadata-driven explanations in improving user understanding and trust in ranking decisions?

1.6 Research Contributions

This thesis makes several contributions to the fields of OGD retrieval, IR, and HCIR.

The first contribution is the development of a fuzzy logic-based retrieval framework specifically designed for OGD environments.

The second contribution is the integration of BM25-based similarity estimation with fuzzy logic-based multi-criteria relevance modeling using similarity, metadata completeness, dataset recency, and resource availability.

The third contribution is the introduction of an explainable ranking mechanism that generates human-readable explanations derived from dataset metadata.

The fourth contribution is a comprehensive evaluation framework that combines quantitative IR benchmarking with qualitative user-centered assessment. Retrieval effectiveness is evaluated by comparing the proposed fuzzy logic-based ranking framework with established baseline retrieval systems using standard IR metrics, while user-centered evaluation investigates usability, transparency, explainability, and user trust through structured participant studies. This mixed-method evaluation provides complementary evidence regarding both ranking performance and human-centered interaction.

Finally, the thesis contributes to HCIR research by showing how fuzzy logic can

support exploratory search, gradual relevance estimation, transparent ranking, and explainable retrieval within Open Government Data environments.

1.7 Thesis Structure

The remainder of this thesis is organized as follows.

Chapter 2 presents the literature review and theoretical background covering OGD, IR, HCIR, BM25 ranking, fuzzy logic, and explainable retrieval systems.

Chapter 3 describes the Design Science Research (DSR) methodology adopted in this study and provides a detailed justification for selecting DSR over alternative research methodologies.

Chapter 4 presents the architecture and design of the proposed retrieval framework, including metadata processing, feature engineering, fuzzy inference, and explainability mechanisms.

Chapter 5 discusses the implementation of the prototype and the technologies used to realize the proposed system.

Chapter 6 presents the complete evaluation methodology adopted for this research. The evaluation consists of two complementary components: a quantitative retrieval evaluation comparing the proposed framework against baseline retrieval approaches using standard IR metrics, and a qualitative user-centered evaluation investigating usability, transparency, explainability, and user trust through participant studies.

Chapter 7 presents and discusses the evaluation results. Quantitative retrieval performance is first analyzed using IR metrics, followed by the statistical analysis of questionnaire responses and qualitative participant feedback. The chapter concludes by discussing the implications of the findings with respect to the proposed research questions, HCIR principles, and the identified limitations of the study.

Finally, Chapter 8 summarizes the contributions of the thesis and outlines future directions for extending the proposed approach.

Chapter 2

Literature Review and Background

This chapter reviews the theoretical foundations and related research relevant to the proposed study. It examines existing work in OGD, IR, HCIR, fuzzy logic, and explainable retrieval systems. The chapter also analyzes the strengths and limitations of existing approaches and identifies the research gap that motivates the development of the proposed fuzzy logic-based retrieval framework.

2.1 Introduction

This chapter presents the theoretical foundations and related research relevant to the development of the proposed HCIR system for Swiss OGD. The purpose of this chapter is to establish the conceptual background of the study, review existing approaches to dataset retrieval, and identify the research gap addressed by this thesis.

The increasing availability of OGD has created significant opportunities for transparency, innovation, and evidence-based decision-making. However, the growing volume of datasets has also introduced challenges related to dataset discoverability and retrieval effectiveness. Users interacting with OGD portals often encounter difficulties when attempting to locate relevant datasets due to inconsistent metadata, vocabulary mismatch, and the exploratory nature of many search tasks [1, 26].

IR research has traditionally focused on keyword-based ranking techniques such as TF-IDF and BM25, which estimate relevance through statistical relationships between query terms and document content [35, 34]. Although these approaches have demonstrated strong performance in many retrieval scenarios, they often struggle to accommodate vague, incomplete, or evolving information needs.

Recent developments in NLP have introduced semantic retrieval methods based on vector embeddings and transformer architectures. Models such as SBERT enable retrieval systems to capture semantic similarity beyond exact keyword matching [33]. However, these approaches often operate as complex black-box systems, making it difficult to ex-

plain why particular results are ranked highly [8].

HCIR emphasizes the importance of designing retrieval systems that support human reasoning, exploratory search behavior, transparency, and trust [26]. Within this context, explainability becomes a critical design requirement, particularly in public-sector information systems where users may require understandable justifications for ranking decisions.

Fuzzy logic offers a promising framework for addressing these challenges. By representing relevance as a gradual rather than binary concept, fuzzy systems can model uncertainty, partial relevance, and subjective interpretations of information needs [42, 21]. Furthermore, fuzzy inference systems provide explicit rule-based reasoning mechanisms that can be translated into human-readable explanations, thereby supporting transparency and interpretability.

This chapter reviews the key concepts underlying OGD, IR, HCIR, fuzzy logic, and explainable retrieval systems. The chapter concludes by identifying limitations in existing approaches and establishing the research gap that motivates the proposed fuzzy logic-based retrieval framework.

2.2 Open Government Data

OGD has emerged as a key component of modern digital government initiatives. By providing public access to government-generated data, OGD promotes transparency, accountability, citizen participation, and innovation. Understanding the principles, benefits, and challenges of OGD is essential because the proposed retrieval system operates within the context of the Swiss national OGD portal, *opendata.swiss*. This section introduces the concept of OGD, discusses its objectives and benefits, and examines the challenges associated with managing and discovering government datasets.

2.2.1 Open Government Data: Concepts, Benefits, and Challenges

OGD refers to data produced, collected, or commissioned by public authorities that is made freely available for public access, use, reuse, and redistribution with minimal legal, technical, or financial restrictions [30, 31]. The objective of OGD initiatives is to increase transparency, encourage citizen participation, stimulate innovation, and improve the accessibility of public-sector information. By making government-generated data openly available, public institutions can support evidence-based decision-making while fostering greater engagement between governments, organizations, researchers, and citizens.

The concept of OGD is closely associated with the broader principles of open government, which emphasize transparency, accountability, and collaboration. According to

the Open Knowledge Foundation, data can be considered open when anyone can freely access, use, modify, and share it subject only to requirements such as attribution and preservation of openness [31]. Governments worldwide have adopted OGD strategies to improve public access to information and to encourage the development of innovative data-driven services and applications.

OGD provides several important benefits. First, it promotes transparency by allowing citizens to access information regarding governmental activities, expenditures, policies, and decision-making processes. Increased transparency can strengthen public trust and improve governmental accountability. Second, OGD supports innovation and economic development by enabling researchers, entrepreneurs, and organizations to create new products, services, and analytical solutions using publicly available data [17]. Third, OGD contributes to evidence-based policymaking by providing access to datasets that support research, policy evaluation, and informed decision-making across multiple domains [17].

Despite these advantages, several challenges continue to limit the effective use of OGD. One of the most significant challenges is dataset discoverability. As the number of published datasets increases, users often struggle to locate relevant information within large repositories containing thousands of datasets [20]. Traditional keyword-based retrieval systems may not adequately support exploratory search activities or complex information needs, particularly when users are unfamiliar with available datasets.

Another challenge concerns metadata quality and consistency. Since datasets are published by different governmental organizations, metadata descriptions often vary in completeness, structure, and terminology. Some datasets contain detailed descriptions, keywords, and documentation, whereas others provide only limited metadata. These inconsistencies can negatively affect retrieval effectiveness and make it difficult for users to evaluate dataset relevance.

Data heterogeneity presents an additional challenge for OGD portals. OGD repositories typically contain datasets from diverse domains such as transportation, health, environment, energy, education, economics, and public administration. These datasets differ substantially in content, format, vocabulary, and update frequency, increasing the complexity of retrieval and ranking processes.

Furthermore, users frequently approach OGD portals with vague, evolving, or exploratory information needs. Rather than searching for a specific known dataset, users often seek to discover information, compare alternatives, or explore unfamiliar topics. Such search behavior introduces uncertainty into the retrieval process and highlights the limitations of retrieval approaches that rely solely on exact keyword matching.

These challenges demonstrate that effective OGD retrieval requires more than conventional search functionality. Retrieval systems must support dataset discoverability, accommodate metadata variability, provide meaningful relevance estimation, and assist users in understanding ranking decisions. These requirements motivate the exploration of

HCIR and fuzzy logic-based approaches, which aim to improve both retrieval effectiveness and user understanding within OGD environments.

2.2.2 Open Government Data in Switzerland

Switzerland supports OGD through the national portal *opendata.swiss*, which serves as a centralized platform for accessing datasets published by federal, cantonal, and municipal institutions [10]. The portal provides access to thousands of datasets covering diverse domains such as transportation, environment, energy, health, population, economy, and public administration.

The portal is built on the CKAN platform and follows the DCAT-AP CH metadata standard, enabling standardized dataset representation and interoperability across organizations. Users can access datasets through both a web interface and APIs, facilitating automated data retrieval and integration.

The large volume of datasets and the diversity of metadata descriptions available on *opendata.swiss* make it an appropriate environment for studying dataset discoverability and evaluating advanced retrieval approaches. Consequently, the portal serves as the primary data source for the retrieval framework developed in this thesis.

2.3 Dataset Discoverability in Open Government Data Portals

The success of OGD initiatives depends not only on the publication of datasets but also on users' ability to discover and access relevant information efficiently. As OGD repositories continue to grow in size and diversity, dataset discoverability has become a major challenge for governments, researchers, and information system designers. Effective discoverability enables users to locate relevant datasets quickly, understand their content, and assess their suitability for specific information needs.

Dataset discoverability is closely linked to metadata quality, retrieval effectiveness, and user interaction. Since users often rely on metadata rather than dataset contents during search, retrieval systems must support efficient metadata exploration while accommodating diverse search behaviors and varying levels of user expertise.

2.3.1 Metadata-Based Retrieval and Discoverability Challenges

Most OGD portals rely on metadata-based retrieval mechanisms [20]. Instead of searching the contents of datasets directly, retrieval systems typically index metadata attributes such as titles, descriptions, keywords, categories, publishers, and update information.

Metadata therefore serves as the primary source of information used to estimate dataset relevance and support user search activities.

Metadata-based retrieval offers several advantages. It enables efficient indexing of large repositories, reduces computational complexity, and provides descriptive information that helps users evaluate datasets before downloading or analyzing them. Consequently, metadata plays a critical role in supporting dataset discovery and information access within OGD environments [28].

However, the effectiveness of metadata-based retrieval depends heavily on metadata quality. In many OGD repositories, metadata is created by multiple organizations using different standards, vocabularies, and levels of detail. As a result, metadata records often vary considerably in completeness, consistency, and descriptive quality.

Incomplete metadata represents a particularly significant challenge. Datasets lacking meaningful descriptions, keywords, or update information may be difficult to retrieve even when they are highly relevant to user queries. Similarly, inconsistent terminology can reduce retrieval effectiveness by preventing accurate matching between user information needs and dataset descriptions.

Another challenge arises from the scale and heterogeneity of OGD repositories. Modern portals frequently contain thousands of datasets spanning diverse domains such as transportation, health, environment, energy, demographics, and public administration. The diversity of topics and metadata structures increases the complexity of retrieval and ranking processes.

These challenges highlight the need for retrieval approaches that extend beyond traditional keyword matching and incorporate additional indicators of relevance, quality, and usefulness.

2.3.2 Exploratory Search in Open Government Data Environments

Unlike traditional lookup searches, exploratory search occurs when users seek to learn, investigate, compare, or discover information rather than retrieve a specific known item [26]. Exploratory search is particularly common within OGD environments because users are often unfamiliar with available datasets and may not know precisely what information exists within the repository.

In many cases, users begin with broad or vague information needs that evolve throughout the search process. They may refine queries, examine multiple datasets, compare metadata descriptions, and gradually develop a better understanding of available information resources [26]. This iterative behavior differs substantially from conventional keyword-based retrieval scenarios.

Supporting exploratory search requires retrieval systems that accommodate uncer-

tainty and provide meaningful guidance during information seeking. Users benefit from ranking mechanisms that consider multiple relevance factors rather than relying exclusively on lexical similarity. Furthermore, transparency and explainability become increasingly important because users often need to understand why datasets were recommended and how ranking decisions were made.

These requirements align closely with the principles of HCIR, which emphasize user understanding, interaction, transparency, and support for complex information-seeking behavior. Consequently, exploratory search provides a strong motivation for developing retrieval frameworks that combine effective ranking mechanisms with explainable and user-centered design principles.

2.4 Information Retrieval Models

IR systems are designed to identify and rank documents according to their relevance to user information needs. Within OGD environments, retrieval systems play an important role in helping users locate datasets from large and heterogeneous repositories. Effective retrieval models must therefore support accurate relevance estimation while remaining computationally efficient and interpretable.

Over the years, numerous retrieval approaches have been proposed, ranging from traditional lexical matching methods to modern semantic retrieval techniques based on deep learning and language models. Understanding the strengths and limitations of these approaches is important for selecting an appropriate retrieval foundation for HCIR systems.

2.4.1 Traditional Information Retrieval Models

Traditional IR approaches estimate relevance primarily through lexical matching between query terms and document content. These methods assume that documents containing terms similar to those used in a query are more likely to satisfy user information needs.

One of the earliest and most influential approaches is Term Frequency–Inverse Document Frequency (TF-IDF), which assigns weights to terms based on their frequency within a document and their rarity across the collection [35]. Although TF-IDF remains widely used, it does not explicitly account for document length normalization or term frequency saturation.

To address these limitations, probabilistic retrieval models such as BM25 were developed. BM25 extends traditional term-weighting approaches by incorporating inverse document frequency, term frequency saturation, and document length normalization. As a result, BM25 has become one of the most widely adopted retrieval models in both academic research and real-world search systems [34].

The BM25 ranking function is defined as:

$$BM25(D, Q) = \sum_{i=1}^n IDF(q_i) \cdot \frac{f(q_i, D)(k_1 + 1)}{f(q_i, D) + k_1 \left(1 - b + b \frac{|D|}{avgdl}\right)} \quad (2.1)$$

where $f(q_i, D)$ represents the frequency of query term q_i in document D , $|D|$ denotes document length, $avgdl$ is the average document length within the collection, and k_1 and b are model parameters controlling term frequency saturation and document length normalization.

Due to its effectiveness, simplicity, and interpretability, BM25 continues to serve as a strong baseline for many retrieval applications.

2.4.2 Semantic Retrieval Models

Recent advances in NLP have led to the development of semantic retrieval approaches that represent documents and queries using dense vector embeddings. Unlike traditional lexical methods, semantic retrieval models attempt to capture contextual meaning and conceptual relationships between terms.

Transformer-based models such as SBERT generate dense vector representations that can be compared using similarity measures to identify semantically related documents [33]. These approaches can improve retrieval performance when relevant documents use different terminology than the submitted query.

Despite their advantages, semantic retrieval models present several challenges. They typically require substantial computational resources, large pre-trained models, and additional infrastructure for vector indexing and similarity search. Furthermore, the reasoning process underlying retrieval decisions is often less transparent than traditional lexical approaches, making explainability more difficult.

For applications emphasizing transparency, interpretability, and explainable decision-making, traditional retrieval models remain attractive alternatives.

2.4.3 Selection of BM25 for the Proposed Framework

Several retrieval approaches were considered during the design of the proposed framework, including TF-IDF, cosine similarity-based retrieval, dense embeddings, and transformer-based semantic retrieval models. However, BM25 was ultimately selected as the primary retrieval mechanism due to its effectiveness, interpretability, and compatibility with HCIR objectives.

Compared with TF-IDF, BM25 provides more sophisticated relevance estimation through document length normalization and term frequency saturation. These mechanisms reduce biases toward longer documents and generally produce more robust ranking

performance [34].

Dense embedding approaches and transformer-based models such as SBERT offer strong semantic capabilities but introduce additional computational complexity and reduced interpretability. Since one of the primary objectives of this research is to support transparency and explainability, highly complex black-box retrieval models were considered less suitable for the proposed framework.

Another important consideration was integration with fuzzy logic-based ranking. BM25 produces interpretable relevance scores that can be incorporated directly into a fuzzy inference system and explained to users. This compatibility supports HCIR principles by enabling transparent relevance estimation and understandable ranking decisions.

Cosine similarity was also considered as a potential retrieval approach. However, cosine similarity primarily measures the angular similarity between vector representations and does not inherently account for document length normalization, term frequency saturation, or probabilistic relevance estimation. In contrast, BM25 was specifically designed for IR applications and has consistently demonstrated strong performance across benchmark retrieval tasks. Furthermore, BM25 provides interpretable relevance scores that can be integrated directly into the fuzzy inference system developed in this research. For these reasons, BM25 was considered more suitable than cosine similarity for metadata-based dataset retrieval within OGD environments [34].

Although recent neural retrieval methods such as Dense Passage Retrieval (DPR), ColBERT, and Large Language Model (LLM)-based rerankers have demonstrated strong retrieval performance, they generally require substantially larger training corpora, greater computational resources, and often provide limited transparency regarding ranking decisions [18, 19, 22, 37]. In contrast, the objective of this research is not solely to maximise retrieval effectiveness but to investigate how multiple metadata-derived relevance criteria can be integrated within an interpretable HCIR framework. BM25 therefore provides an appropriate lexical retrieval foundation because it is computationally efficient, reproducible, and widely adopted as a strong baseline in IR research. Its transparent scoring mechanism also allows the subsequent fuzzy inference layer to explain how different metadata characteristics contribute to the final ranking, which would be considerably more difficult with many neural retrieval approaches.

2.5 HCIR

Traditional IR research has primarily focused on improving ranking effectiveness and retrieval accuracy. While these objectives remain important, modern retrieval systems must also consider how users interact with information systems, interpret retrieval outcomes, and make decisions based on retrieved information.

HCIR extends traditional retrieval approaches by placing users at the center of the

retrieval process. Rather than treating retrieval as a purely algorithmic task, HCIR emphasizes interaction, transparency, interpretability, and support for human decision-making. The objective is not only to retrieve relevant information but also to help users understand, explore, and effectively utilize retrieval results.

2.5.1 Principles of HCIR

HCIR is founded on the recognition that information seeking is an interactive and evolving process [36, 26]. Users frequently refine queries, explore alternative information sources, and modify search objectives as their understanding develops. Consequently, retrieval systems should support exploration and learning rather than merely returning ranked lists of documents.

Several principles characterize HCIR. First, retrieval systems should support user interaction by enabling exploration, query refinement, and iterative search behavior. Second, systems should provide meaningful feedback that helps users understand search results and navigate information spaces effectively. Third, retrieval mechanisms should accommodate uncertainty because users often begin searches with incomplete or evolving information needs.

Unlike purely system-centered approaches, HCIR recognizes that successful retrieval depends not only on algorithmic performance but also on user understanding, satisfaction, and trust. These principles are particularly relevant within OGD environments where users frequently engage in exploratory search activities and may possess varying levels of technical expertise.

2.5.2 Transparency and Explainability in HCIR

Transparency and explainability represent fundamental objectives of HCIR [36]. Users are more likely to trust retrieval systems when they can understand how ranking decisions are generated and why particular results are recommended.

Traditional retrieval systems often provide limited insight into the reasoning process underlying ranking outcomes. As a result, users may find it difficult to evaluate the reliability or relevance of retrieved information. Explainable retrieval systems address this challenge by providing understandable descriptions of ranking decisions and relevance estimation processes.

Within HCIR, explainability supports several important objectives. First, it improves user understanding by making retrieval processes more accessible. Second, it enhances trust by allowing users to assess the reasoning behind recommendations. Third, it supports learning and exploration by helping users understand relationships between queries, metadata attributes, and retrieval outcomes.

Consequently, transparency and explainability are increasingly recognized as essential characteristics of user-centered retrieval systems.

2.5.3 HCIR in Open Government Data Environments

The principles of HCIR are particularly relevant within OGD environments [20]. Users of OGD portals often possess diverse backgrounds, varying levels of technical expertise, and different information needs. Furthermore, many users engage in exploratory search activities and may be unfamiliar with available datasets or domain-specific terminology.

Traditional keyword-based retrieval systems may struggle to support these complex information-seeking behaviors because they focus primarily on lexical matching rather than user understanding. HCIR approaches seek to address these limitations by emphasizing interaction, transparency, explainability, and support for exploratory search.

In the context of this research, HCIR provides the conceptual foundation for the development of an explainable retrieval framework that combines BM25 retrieval, fuzzy logic-based ranking, and metadata-driven explanations. By integrating these components, the proposed framework aims to support transparent and interpretable retrieval while maintaining competitive retrieval effectiveness within the Swiss OGD portal.

2.6 Fuzzy Logic and Information Retrieval

Traditional retrieval systems frequently represent relevance as a binary concept, where documents are considered either relevant or non-relevant. However, human judgments rarely operate in such a strictly categorical manner. Users often perceive relevance as gradual, uncertain, and context-dependent. Consequently, alternative approaches are required to model the ambiguity and subjectivity inherent in information-seeking behavior.

Fuzzy logic provides a mathematical framework for representing uncertainty and gradual reasoning. Introduced by Zadeh [42], fuzzy logic allows concepts to be represented through varying degrees of membership rather than binary classifications. This capability makes fuzzy logic particularly attractive for IR applications where relevance is often subjective and difficult to define precisely.

2.6.1 Fundamentals of Fuzzy Logic

Unlike classical Boolean logic, which assigns elements exclusively to either membership or non-membership categories, fuzzy logic allows partial membership. Membership values typically range between 0 and 1, representing the degree to which an element belongs to a particular fuzzy set [42].

Fuzzy logic utilizes linguistic variables such as Low, Medium, and High to describe concepts that cannot be defined precisely. Membership functions are used to map nu-

merical values into these linguistic categories, thereby enabling gradual reasoning and the representation of uncertainty [42].

This flexibility allows fuzzy systems to model complex human judgments more naturally than traditional binary approaches. Consequently, fuzzy logic has been widely applied in decision support systems, expert systems, control systems, and IR applications.

2.6.2 Fuzzy Inference Systems

A fuzzy inference system combines fuzzy sets, linguistic variables, and rule-based reasoning to generate decisions or relevance estimates. The process typically consists of four stages: fuzzification, rule evaluation, aggregation, and defuzzification.

Among the available inference approaches, the Mamdani model is one of the most widely used because of its interpretability and compatibility with human reasoning [24]. Mamdani systems employ IF–THEN rules that describe relationships between input variables and output decisions using linguistic terms.

For example:

IF Similarity is High AND Metadata Completeness is High THEN Relevance is Very High.

Such rules resemble human decision-making processes and therefore contribute to transparency and explainability.

2.6.3 Fuzzy Logic in Information Retrieval

Fuzzy logic has been applied extensively within IR research to address limitations associated with binary relevance judgments [4]. Rather than classifying documents as simply relevant or non-relevant, fuzzy retrieval approaches represent relevance as a gradual concept characterized by varying degrees of membership.

Several studies have reported that fuzzy logic can support retrieval by integrating multiple relevance criteria and accommodating uncertainty within user information needs. Fuzzy approaches are particularly useful in environments where relevance depends on multiple factors rather than a single similarity measure.

Within OGD repositories, dataset relevance may depend not only on textual similarity but also on metadata quality, recency, completeness, and contextual suitability. Fuzzy inference systems provide a natural mechanism for combining these heterogeneous relevance indicators into a unified ranking framework.

2.6.4 Why Fuzzy Logic Supports HCIR

One of the central arguments of this thesis is that fuzzy logic provides characteristics that align closely with HCIR principles.

First, fuzzy logic models gradual relevance rather than binary relevance [4]. Human users rarely perceive information as completely relevant or completely irrelevant. Instead, relevance is often evaluated along a continuum influenced by context, task requirements, and personal preferences. Fuzzy logic naturally supports this type of reasoning.

Second, fuzzy inference systems accommodate uncertainty and ambiguity [42]. Users frequently submit incomplete, vague, or exploratory queries, particularly within OGD environments. Fuzzy reasoning enables retrieval systems to handle such uncertainty more effectively than rigid rule-based approaches.

Third, fuzzy logic promotes transparency and explainability through explicit IF-THEN rules [36]. Unlike many black-box machine learning models, fuzzy systems allow users to understand how relevance estimates are generated and which factors contribute to ranking decisions.

Finally, fuzzy logic facilitates the integration of multiple relevance indicators. In the proposed framework, thematic similarity, metadata completeness, and dataset recency are combined through fuzzy reasoning to generate relevance estimates that more closely reflect human judgment.

For these reasons, fuzzy logic represents a suitable foundation for HCIR systems that emphasize transparency, interpretability, and user understanding.

From a HCIR perspective, fuzzy logic enables retrieval systems to operate in a manner that more closely resembles human reasoning. Human users often evaluate relevance using imprecise concepts such as "mostly relevant", "somewhat useful", or "highly related" rather than strict numerical thresholds. Fuzzy logic allows these gradual judgments to be represented mathematically through linguistic variables and membership functions. Consequently, fuzzy reasoning provides a natural bridge between computational ranking mechanisms and human relevance perception, making it particularly suitable for HCIR environments [3, 42].

While alternative approaches such as weighted-sum ranking, learning-to-rank models, and multi-criteria decision-making methods (e.g., AHP or TOPSIS) can also combine multiple relevance factors, they present different trade-offs. Weighted-sum approaches assume fixed linear relationships between ranking criteria and cannot naturally represent gradual transitions between linguistic concepts such as *low*, *medium*, and *high* relevance. Learning-to-rank methods generally require substantially larger collections of labelled training data than were available in this study and often reduce transparency by producing complex learned ranking functions. Similarly, multi-criteria decision-making methods are well suited for structured decision problems but typically provide less flexibility for

modelling uncertainty and subjective relevance judgments. Fuzzy inference was therefore selected because it supports interpretable rule-based reasoning while allowing multiple metadata-derived relevance criteria to be combined in a manner that closely reflects human reasoning under uncertainty.

Among the available fuzzy inference approaches, the Mamdani model was selected in preference to Sugeno or Takagi–Sugeno–Kang (TSK) inference because the primary objective of this research is interpretability rather than computational optimisation. Mamdani inference produces linguistic output membership functions that can be directly translated into human-readable explanations, thereby supporting transparency and explainability within HCIR. Although Sugeno-based systems are computationally efficient and are often preferred for optimisation and control applications, their numerical outputs are generally less intuitive for explaining ranking decisions to end users. Consequently, Mamdani inference provides a more appropriate foundation for the proposed explainable retrieval framework [38, 42].

2.7 Explainable Information Retrieval

As retrieval systems become increasingly complex, users often find it difficult to understand why particular documents or datasets are recommended. Although advanced ranking algorithms may improve retrieval effectiveness, a lack of transparency can reduce user trust and hinder effective decision-making. Consequently, explainability has emerged as an important research area within IR.

Explainable Information Retrieval (XIR) seeks to provide users with understandable descriptions of ranking decisions and retrieval outcomes. Rather than presenting search results as opaque recommendations, explainable systems communicate the factors that influenced relevance estimation and document ranking. This transparency helps users interpret retrieval results and assess their suitability for specific information needs.

2.7.1 Explainability in Information Retrieval

Explainability refers to the ability of a system to communicate the reasoning process underlying its decisions in a manner that is understandable to human users [27]. Within IR, explainability aims to clarify why particular documents are retrieved, how relevance scores are generated, and which factors contribute to ranking decisions.

Explainable retrieval systems can provide information regarding query-document similarity, metadata characteristics, relevance indicators, and ranking criteria. Such explanations help users understand retrieval outcomes and improve confidence in system recommendations.

2.7.2 Benefits and Challenges of Explainability

Explainability provides several important benefits within retrieval systems [27]. First, it improves transparency by exposing the factors that influence ranking decisions. Second, it supports user trust by enabling users to understand and evaluate retrieval outcomes. Third, explanations can facilitate learning and exploratory search by helping users understand relationships between queries and retrieved information.

Despite these advantages, implementing explainability presents several challenges. Explanations must remain accurate while also being understandable to users with varying levels of expertise. Excessively technical explanations may confuse users, whereas overly simplified explanations may fail to communicate meaningful information.

Designing effective explanations therefore requires balancing transparency, usability, and informational value. HCIR emphasizes this balance by focusing on explanations that support user understanding and decision-making.

2.7.3 Explainability for Open Government Data Retrieval

Explainability is particularly valuable within OGD environments because users often possess limited knowledge regarding available datasets, metadata structures, and retrieval mechanisms. Consequently, users may find it difficult to determine why specific datasets are recommended or how relevance scores are generated.

Providing explanations alongside retrieval results can improve user understanding of dataset characteristics and assist users in evaluating dataset suitability [27]. Metadata-driven explanations are especially useful because they communicate observable factors such as thematic similarity, metadata completeness, and dataset recency.

In the proposed framework, explainability is achieved through the integration of fuzzy logic-based reasoning and metadata-driven relevance estimation. The use of explicit fuzzy rules enables ranking decisions to be communicated in a transparent and understandable manner, thereby supporting HCIR objectives and improving user trust in retrieval outcomes.

Unlike post-hoc explanation methods that generate explanations independently of the underlying ranking process, the proposed framework produces *faithful* explanations that directly reflect the reasoning performed by the fuzzy inference system. The generated explanations are derived from the same metadata features, fuzzy membership functions, and rule activations used to compute the final relevance score. Consequently, the explanation presented to the user corresponds directly to the actual ranking process rather than providing a separate or approximate interpretation. This close correspondence between ranking and explanation supports transparency, accountability, and user trust, which are fundamental objectives of HCIR and Explainable IR.

2.8 Research Gap and Conceptual Framework

The literature reviewed in this chapter demonstrates significant progress in OGD retrieval, IR, HCIR, fuzzy logic, and explainable retrieval systems. However, several limitations remain that motivate the development of the proposed framework.

Although existing studies provide valuable contributions, relatively few approaches integrate retrieval effectiveness, explainability, transparency, and user-centered design within a unified OGD retrieval framework. Furthermore, limited research has explored the combined use of BM25 retrieval, fuzzy logic-based relevance estimation, and HCIR principles for dataset discovery. These observations form the basis for the research gap addressed by this thesis.

2.8.1 Research Gap

The literature reveals several limitations in current OGD retrieval systems.

First, many OGD portals rely primarily on traditional keyword-based retrieval mechanisms. While such approaches can efficiently identify datasets containing matching query terms, they often struggle to accommodate exploratory search behavior, uncertainty, and evolving information needs. Consequently, retrieval effectiveness may be limited when users are unfamiliar with available datasets or employ vocabulary different from that used in metadata descriptions.

Second, existing retrieval approaches frequently focus on textual similarity while overlooking additional factors that may influence dataset relevance. Characteristics such as metadata completeness, dataset recency, and metadata quality can affect the usefulness of datasets but are often excluded from ranking processes.

Third, explainability remains limited within many retrieval environments. Users are typically presented with ranked results without receiving meaningful explanations regarding how relevance scores were generated. This lack of transparency can reduce trust and make it difficult for users to evaluate retrieval outcomes.

Finally, despite increasing interest in HCIR, relatively little research has investigated the integration of HCIR principles, fuzzy logic-based reasoning, and OGD retrieval within a unified framework. Existing studies tend to focus on individual aspects of retrieval rather than providing a comprehensive solution that combines retrieval effectiveness, transparency, explainability, and user-centered interaction.

These limitations highlight the need for retrieval frameworks capable of supporting both effective dataset discovery and meaningful user understanding.

More specifically, no identified study was found that combines BM25 retrieval, metadata completeness assessment, dataset recency evaluation, fuzzy logic-based relevance estimation, and explainable HCIR within the context of the Swiss OGD portal [34, 4, 36].

This specific combination constitutes the primary research gap addressed by the present study.

2.8.2 Conceptual Framework

To address the identified research gap, this thesis proposes a HCIR framework for Swiss OGD. The framework integrates BM25-based retrieval, fuzzy logic-based relevance estimation, metadata-driven ranking, explainability mechanisms, and a user-centered interface within a unified retrieval architecture [34, 42, 3].

The framework is designed to improve dataset discoverability while supporting transparency, explainability, and user understanding. Instead of relying solely on textual similarity, the framework combines multiple relevance indicators, including thematic similarity, metadata completeness, dataset recency, and resource availability to produce relevance estimates that more closely reflect human judgment.

Table 2.1 summarizes the major components of the proposed framework and their respective roles.

Table 2.1: Conceptual Framework Components

Component	Purpose
Metadata Acquisition and Processing	Collects and preprocesses metadata from the Swiss OGD portal.
BM25 Retrieval Engine	Calculates thematic similarity between user queries and dataset metadata.
Fuzzy Inference System	Combines thematic similarity, metadata completeness, and dataset recency and resource availability to estimate relevance.
Explainability Module	Generates understandable explanations for ranking decisions.
User Interface (UI)	Supports user interaction, dataset exploration, and result visualization.

The interaction among these components enables the framework to move beyond traditional keyword-based retrieval approaches. By integrating retrieval effectiveness with explainability and user-centered design principles, the proposed framework aims to improve dataset discovery and support more effective interaction with OGD resources.

2.9 Chapter Summary

This chapter reviewed the literature related to OGD, dataset discoverability, IR, HCIR, fuzzy logic, and explainable retrieval systems. The review identified limitations in existing OGD retrieval approaches, particularly regarding transparency, explainability, and support for exploratory search. Based on these observations, a HCIR framework integrating BM25 retrieval, fuzzy logic-based ranking, and explainability mechanisms was proposed to address the identified research gap.

Chapter 3

Research Methodology

This chapter describes the research methodology adopted to guide the design, development, and evaluation of the proposed retrieval framework. The chapter introduces the DSR paradigm, explains its suitability for the objectives of this study, and presents the research process followed throughout the project. In addition, the chapter discusses alternative research methodologies and justifies the selection of DSR as the most appropriate methodological approach for developing and evaluating a HCIR system for Swiss OGD.

3.1 Introduction to Research Methodology

Research methodology provides the systematic framework through which scientific investigations are conducted. The choice of an appropriate methodology is critical because it influences how research questions are addressed, how data is collected and analyzed, and how research contributions are evaluated. Different methodologies are suitable for different research objectives, ranging from theory development and hypothesis testing to artifact design and practical problem solving.

The objective of this thesis is not only to study an existing phenomenon but also to design, implement, and evaluate a novel retrieval framework that addresses identified limitations in OGD retrieval systems. Consequently, a methodology capable of supporting artifact creation, iterative refinement, and practical evaluation is required. This requirement motivated the adoption of DSR as the primary methodological framework for the study.

3.2 Design Science Research

DSR is a research paradigm that focuses on the creation and evaluation of innovative artifacts designed to solve identified problems [13]. Unlike purely descriptive research approaches, DSR aims to generate practical solutions while simultaneously contributing

to scientific knowledge.

In Information Systems research, artifacts may take various forms, including models, frameworks, methods, algorithms, software systems, and design principles. The central objective of DSR is to develop artifacts that address real-world problems while providing insights that extend beyond the specific implementation.

According to Hevner et al. [13], DSR consists of iterative cycles involving problem identification, artifact design, implementation, evaluation, and refinement. Through these cycles, researchers continuously improve artifact quality while generating new knowledge regarding its effectiveness and applicability.

The proposed retrieval framework developed in this thesis constitutes a design artifact because it introduces a new combination of BM25 retrieval, fuzzy inference-based ranking, and explainable relevance estimation for OGD retrieval. Therefore, DSR provides an appropriate methodological foundation for guiding the development and evaluation process.

3.3 Why Design Science Research?

The selection of DSR was motivated by the objectives and nature of this study. The primary goal of the research is to design and evaluate a novel retrieval framework rather than merely observe or explain an existing phenomenon. Consequently, a methodology focused on artifact creation and evaluation is required.

First, DSR supports the development of innovative solutions to practical problems. The research addresses the challenge of dataset discoverability within OGD portals through the design of a fuzzy logic-based HCIR framework. Since the study seeks to produce a functional retrieval system, DSR aligns naturally with the research objectives.

Second, DSR encourages iterative development and refinement. The proposed retrieval framework underwent multiple stages of design, implementation, evaluation, and improvement throughout the project. This iterative process is consistent with the principles of DSR and allows continuous enhancement of artifact quality.

Third, DSR emphasizes both practical relevance and scientific rigor. The research contributes practically by improving dataset retrieval and contributes theoretically by demonstrating how fuzzy logic can support HCIR principles within OGD environments.

Finally, DSR is widely accepted within Information Systems research and has been successfully applied in studies involving decision support systems, recommendation systems, retrieval systems, and HCI. Therefore, DSR provides a well-established methodological framework for the development of the proposed retrieval artifact [13, 32].

3.3.1 Justification for Selecting Design Science Research

Several research methodologies were considered during the planning of this study, including experimental research, survey research, case study research, and DSR. Design Science Research was ultimately selected because the primary objective of the research was not only to study an existing phenomenon but also to design, develop, and evaluate a novel artifact capable of addressing an identified problem.

The central problem addressed by this research concerns dataset discoverability within the Swiss OGD portal. Existing retrieval approaches often provide limited support for exploratory search, explainability, and user-centered interaction. To address these limitations, this research proposes and implements a HCIR framework that combines BM25 retrieval, fuzzy logic-based relevance estimation, and explainability mechanisms.

Unlike purely observational methodologies, DSR explicitly focuses on the creation and evaluation of innovative artifacts that solve real-world problems [32]. The retrieval framework developed in this study constitutes such an artifact. Consequently, DSR provides a structured methodology for guiding artifact design, implementation, demonstration, and evaluation [32].

Alternative methodologies were considered less suitable for the objectives of this research. Experimental research primarily focuses on testing hypotheses rather than designing artifacts. Survey research is appropriate for collecting perceptions and opinions but does not provide guidance for system development. Case study research enables in-depth investigation of existing environments but does not emphasize artifact construction. In contrast, DSR directly supports the iterative design and evaluation of information systems intended to address practical challenges.

Therefore, DSR was selected because it provides a rigorous and widely accepted framework for developing and evaluating innovative information system artifacts while simultaneously contributing to scientific knowledge and practical problem solving.

3.4 Why Not Other Research Methodologies?

Several alternative research methodologies were considered before selecting DSR. Although these methodologies offer valuable perspectives, they were not fully aligned with the objectives of the present study.

3.4.1 Survey Research

Survey research focuses primarily on collecting opinions, attitudes, and perceptions from participants through questionnaires and interviews. While surveys are useful for understanding user behavior and preferences, they do not provide a structured framework for designing and evaluating technological artifacts.

Since the primary objective of this research is to develop and evaluate a retrieval system rather than solely investigate user perceptions, survey research alone would be insufficient. Nevertheless, survey techniques were employed during the evaluation phase to assess user satisfaction and usability.

3.4.2 Experimental Research

Experimental research is commonly used to test hypotheses by manipulating variables under controlled conditions. Although experimental methods can evaluate system performance, they do not provide guidance for artifact design and development.

The present study involves the creation of a retrieval framework rather than the testing of isolated causal relationships. Consequently, experimental research was incorporated as part of the evaluation process but was not selected as the overall methodological framework.

3.4.3 Case Study Research

Case study research aims to investigate phenomena within specific real-world contexts. While case studies provide rich contextual insights, they generally focus on understanding existing situations rather than creating new technological solutions.

The goal of this research is to develop and evaluate a novel retrieval framework rather than analyze a particular organizational setting. Therefore, case study research was not considered the most appropriate primary methodology.

3.4.4 Action Research

Action Research involves close collaboration between researchers and practitioners to solve organizational problems through cycles of intervention and reflection. Although Action Research supports practical problem solving, it typically requires continuous organizational participation throughout the research process.

The proposed retrieval framework was developed independently and evaluated through controlled user studies rather than organizational interventions. Consequently, DSR provided a more suitable methodological approach.

3.5 Design Science Research Process

The research process followed the DSR framework proposed by Hevner et al. [13]. The process consisted of five major phases:

1. Problem Identification

2. Objectives Definition
3. Artifact Design
4. Artifact Development
5. Evaluation and Refinement

The research process began with an analysis of dataset discoverability challenges within OGD portals. Following the identification of research gaps, research objectives were formulated to guide the design of a HCIR framework.

The retrieval artifact was subsequently designed and implemented using a custom BM25 retrieval component, a Mamdani fuzzy inference engine, and metadata-driven explanation mechanisms. The completed prototype was evaluated using a mixed-method evaluation strategy consisting of quantitative retrieval benchmarking and qualitative user-centered assessment. Quantitative evaluation compared the proposed framework with baseline retrieval approaches using standard IR metrics, while the qualitative evaluation examined usability, transparency, explainability, and user trust through structured participant studies. The combined findings were then used to assess the proposed framework and identify opportunities for further refinement.

3.5.1 Research Process Overview

The overall research process adopted in this study consisted of the following stages:

1. Literature Review
2. Problem Identification
3. Research Gap Analysis
4. Artifact Design
5. Prototype Development
6. System Evaluation
7. Refinement and Improvement
8. Final Analysis and Conclusions

The iterative nature of this process ensured that evaluation results continuously informed artifact improvements, thereby increasing both system quality and research rigor.

3.6 Chapter Summary

This chapter presented the research methodology adopted for the study and justified the selection of DSR as the guiding methodological framework. DSR was chosen because it supports the design, implementation, evaluation, and refinement of technological artifacts while maintaining scientific rigor and practical relevance.

The chapter also discussed alternative methodologies and explained why they were less suitable for achieving the objectives of the study. Finally, the DSR process followed throughout the project was presented. The next chapter describes the architecture and design of the proposed fuzzy logic-based HCIR framework.

Chapter 4

System Design

This chapter presents the architecture and design of the proposed HCIR framework for Swiss OGD. The chapter describes the overall system architecture, data flow, retrieval components, fuzzy inference framework, and explainability mechanisms. Particular attention is given to the integration of BM25 retrieval and fuzzy logic-based ranking, which together form the core of the proposed retrieval system.

4.1 Introduction

The design of an effective retrieval system requires the integration of multiple components that collectively support dataset discovery, relevance estimation, and user interaction. Based on the limitations identified in existing OGD retrieval systems, the proposed framework was designed to support exploratory search, transparent ranking, and explainable decision-making.

The system combines traditional IR techniques with fuzzy logic-based reasoning in order to provide a more human-centered retrieval experience. Rather than relying solely on keyword matching, the framework incorporates multiple metadata-driven relevance factors and produces interpretable explanations for ranking decisions. This section presents the overall design principles and architectural considerations that guided the development of the system.

4.2 Design Objectives

The design of the proposed retrieval framework was guided by several objectives derived from the research questions and literature review.

First, the system should improve dataset discoverability within OGD environments. Users should be able to locate relevant datasets efficiently even when queries are vague or exploratory.

Second, the framework should support HCIR principles by providing transparency, explainability, and user-friendly interaction mechanisms.

Third, the system should integrate multiple relevance criteria rather than relying solely on keyword similarity. Metadata completeness, dataset recency, and retrieval relevance should all contribute to ranking decisions.

Fourth, the framework should remain computationally efficient and scalable for practical deployment within OGD portals.

Finally, the retrieval process should be explainable so that users can understand why datasets are ranked in particular positions.

4.3 Overall System Architecture

The proposed system follows a layered architecture consisting of metadata acquisition, metadata processing, retrieval, fuzzy ranking, explainability, and user interaction components. User queries are processed through the retrieval layer, which estimates thematic similarity using BM25. Additional metadata quality indicators are then evaluated through a fuzzy inference system to generate final relevance scores. The explainability layer produces human-readable explanations that are presented through the UI. The proposed retrieval framework consists of five major components:

1. Dataset Acquisition Layer
2. Metadata Processing Layer
3. BM25 Retrieval Layer
4. Fuzzy Ranking Layer
5. Explainability and UI Layer

The retrieval process begins when a user submits a search query through the interface. The query is processed and matched against indexed dataset metadata using the BM25 retrieval model. The retrieved datasets are then evaluated using a fuzzy inference system that combines multiple metadata-driven relevance criteria.

The resulting relevance scores are used to generate the final ranking presented to the user. In addition, metadata-based explanations are generated to provide transparency regarding ranking decisions. This architecture supports both retrieval effectiveness and HCIR objectives.

4.4 Dataset Acquisition Layer

The first component of the framework is responsible for acquiring dataset metadata from the Swiss OGD portal, *opendata.swiss*. The portal provides access to datasets through a CKAN-based API, allowing metadata records to be retrieved programmatically.

Metadata fields collected from the portal include:

- Dataset title
- Dataset description
- Publisher information
- Tags and keywords
- Publication date
- Update date
- Dataset category
- Resource information

The acquired metadata forms the foundation of the retrieval process because relevance estimation is performed primarily on metadata rather than the full dataset contents.

4.5 Metadata Processing Layer

Dataset metadata obtained from the Swiss OGD portal undergoes preprocessing before indexing and retrieval. To emphasize the relative importance of metadata attributes, different weighting strategies are applied during document construction.

The title field receives the highest weight and is repeated three times within the document representation. Tags and groups are repeated twice to increase their influence during retrieval. Dataset descriptions are included once, while organizational information is only used as a fallback when structured metadata is unavailable.

This weighted representation enables important metadata attributes to contribute more strongly to retrieval performance while preserving the original metadata content.

The preprocessing pipeline includes:

1. Text normalization
2. Lowercase conversion
3. Removal of punctuation

4. Tokenization
5. Stop-word removal
6. Metadata validation

In addition to textual preprocessing, metadata completeness indicators are calculated. These indicators measure the availability of important metadata attributes and later contribute to fuzzy relevance estimation.

Metadata preprocessing improves retrieval effectiveness by reducing inconsistencies and ensuring that dataset descriptions can be compared more reliably during the retrieval process.

4.6 BM25 Retrieval Component

The first stage of relevance estimation is performed using the BM25 retrieval model. BM25 was selected because it provides robust relevance estimation, document length normalization, and strong retrieval performance across diverse IR tasks.

When a user submits a query, BM25 calculates similarity scores between the query and available dataset metadata. The resulting scores provide an initial ranking of candidate datasets.

The BM25 score is computed using:

$$BM25(D, Q) = \sum_{i=1}^n IDF(q_i) \cdot \frac{f(q_i, D)(k_1 + 1)}{f(q_i, D) + k_1 \left(1 - b + b \frac{|D|}{avgdl}\right)} \quad (4.1)$$

where the variables correspond to query term frequencies, document lengths, and inverse document frequency values.

The BM25 component serves as the retrieval foundation upon which fuzzy relevance estimation is subsequently applied.

4.7 Fuzzy Ranking Framework

While BM25 provides an effective estimate of textual similarity, relevance within OGD environments depends on additional factors. Consequently, a fuzzy inference system was developed to combine multiple metadata-driven relevance criteria into a unified relevance score.

The fuzzy ranking framework evaluates:

- BM25 similarity score
- Metadata completeness

- Dataset recency
- Resource availability

Each input variable is transformed into fuzzy linguistic values using membership functions. The fuzzy inference system then applies predefined rules to estimate overall relevance.

The output of the fuzzy system is a final relevance score that reflects both textual similarity and metadata quality characteristics. This approach enables more flexible and human-centered ranking behavior than traditional keyword-based retrieval alone.

4.8 Fuzzy Membership Functions

Membership functions define how numerical input variables are transformed into linguistic concepts used by the fuzzy inference engine. Depending on the characteristics of each variable, the implementation employs between three and five linguistic terms. For example, similarity is represented using five linguistic categories:

- Not Relevant,
- Somewhat Relevant
- Relevant
- Highly Relevant
- Exact Match

While dataset recency is represented using five temporal categories ranging from Very Recent to Very Old. This richer linguistic representation enables the fuzzy inference system to model gradual transitions in relevance more accurately than a simple three-level categorization.

4.9 Fuzzy Rule Base

The fuzzy rule base represents the core reasoning mechanism of the proposed retrieval framework. The rules combine multiple relevance criteria using IF-THEN statements that resemble human decision-making.

Examples of fuzzy rules include:

- IF BM25 Similarity is High AND Metadata Completeness is High AND Dataset Recency is High AND Resource Availability is High THEN Relevance is Very High.

- IF BM25 Similarity is Medium AND Metadata Completeness is High THEN Relevance is High.
- IF BM25 Similarity is Low THEN Relevance is Low.

The rule base enables the system to incorporate expert knowledge and domain-specific reasoning into the ranking process. Furthermore, the explicit structure of the rules supports transparency and explainability.

4.10 Explainability Component

Explainability is incorporated as a fundamental design requirement within the proposed framework. After relevance scores are generated, the system produces human-readable explanations that describe why datasets were ranked highly.

Explanations are derived from metadata attributes and fuzzy inference outcomes. For example, a highly ranked dataset may receive an explanation indicating that it achieved strong keyword similarity, contained complete metadata, and was recently updated.

The explainability component improves transparency and supports user trust by making ranking decisions understandable. This feature aligns closely with HCIR principles and distinguishes the proposed framework from traditional black-box retrieval systems.

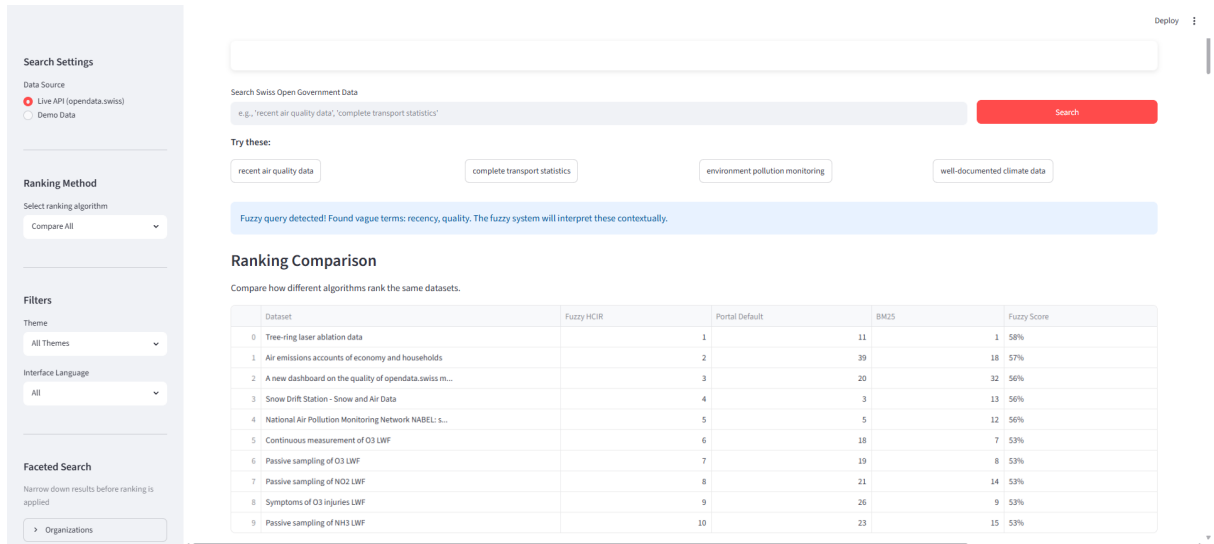


Figure 4.1: Comparison of dataset rankings generated by the proposed Fuzzy HCIR framework, BM25 retrieval, and the default portal ranking mechanism.

Figure 4.1 illustrates the ranking comparison interface developed within the proposed retrieval framework. The interface enables direct comparison between rankings produced by the Fuzzy HCIR approach, the BM25 retrieval model, and the default ranking mechanism provided by the opendata.swiss portal. By presenting ranking positions side by side,

users can observe how the incorporation of metadata completeness and dataset recency influences the final ordering of search results. This functionality improves transparency and allows users to better understand the impact of fuzzy relevance estimation compared with traditional keyword-based retrieval approaches. The comparison view therefore supports the HCIR objective of making ranking decisions more interpretable and understandable to users [3, 27].

4.11 User Interface Design

The UI was developed using Streamlit and serves as the primary interaction point between users and the retrieval system. The interface allows users to submit search queries, explore ranked datasets, and view generated explanations.

Key interface features include:

- Search query input
- Ranked dataset presentation
- Relevance score visualization
- Metadata summaries
- Explanation display

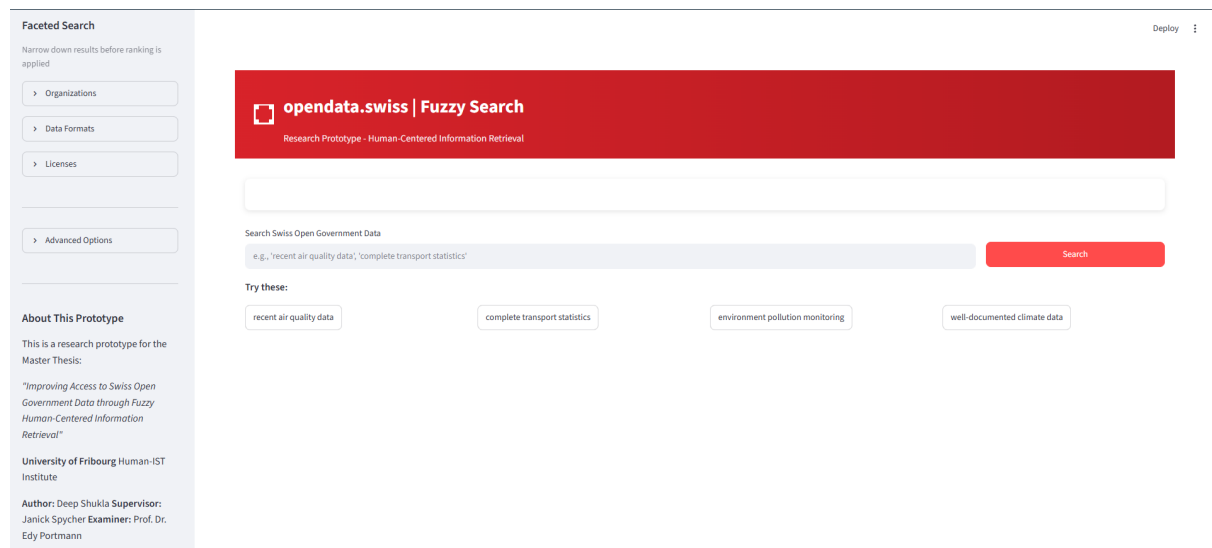


Figure 4.2: Main user interface of the proposed Fuzzy HCIR prototype for Swiss Open Government Data.

Figure 4.2 illustrates the main UI developed for the proposed retrieval framework. The design integrates query submission, faceted navigation, and metadata-based filtering

within a unified environment to support dataset discovery. The central search area enables users to express information needs using natural-language queries, while the faceted search panel provides mechanisms for narrowing result sets before ranking is applied. Additionally, example queries are provided to support exploratory search behaviour and reduce user effort during query formulation. The interface was designed according to HCIR principles, emphasizing usability, transparency, and user control throughout the search process [3, 26].

The interface was designed according to HCIR principles, emphasizing simplicity, transparency, and support for exploratory search activities.

4.12 System Workflow

The overall retrieval workflow can be summarized as follows:

1. User submits query.
2. Metadata is retrieved and preprocessed.
3. BM25 calculates initial similarity scores.
4. Fuzzy inference evaluates multiple relevance criteria.
5. Final relevance scores are generated.
6. Datasets are ranked.
7. Explanations are generated.
8. Results are displayed to the user.

This workflow integrates traditional retrieval techniques with fuzzy reasoning and explainability mechanisms to support HCIR objectives.

4.13 Chapter Summary

This chapter presented the architecture and design of the proposed HCIR framework. The system combines BM25 retrieval, fuzzy logic-based relevance estimation, and explainability mechanisms to support effective dataset discovery within OGD environments. The architecture was designed to balance retrieval effectiveness, transparency, interpretability, and user experience. The next chapter describes the implementation of the prototype and the technologies used to realize the proposed framework.

Chapter 5

Prototype Implementation

This chapter describes the implementation of the proposed HCIR framework. While the previous chapter presented the conceptual design of the system, this chapter focuses on the technologies, software components, and implementation procedures used to realize the proposed architecture. The implementation integrates OGD retrieval, BM25 ranking, fuzzy inference-based relevance estimation, and explainability mechanisms into a functional prototype designed to support dataset discovery.

5.1 Introduction

The implementation phase transforms the conceptual design presented in the previous chapter into a working software artifact. The primary objective of the implementation is to develop a prototype capable of retrieving datasets from the Swiss OGD portal, ranking them using fuzzy logic-based relevance estimation, and providing explainable retrieval results.

The implementation follows the DSR process and emphasizes practicality, transparency, and reproducibility. The resulting prototype serves as the evaluation artifact through which the research questions are investigated and validated.

5.2 Development Environment and Technology Stack

The prototype was implemented using Python and supporting open-source libraries selected according to the functional requirements of the system. Table 5.1 summarizes the primary technologies used during development.

Table 5.1: Technology Stack

Component	Technology
Programming Language	Python
User Interface	Streamlit
Data Processing	Pandas, NumPy
Scientific Computing	SciPy
Visualization	Matplotlib, Seaborn
API Communication	Requests
PDF Processing	PyPDF2
Testing	Pytest, Pytest-Cov
Code Quality	Black, Flake8, MyPy

5.2.1 Custom Implementation Strategy

A deliberate design decision was made to implement the BM25 retrieval model and fuzzy inference engine from scratch rather than relying on external libraries. Although several mature implementations are available, custom development provided greater control over the retrieval pipeline and allowed complete visibility into each stage of relevance estimation.

The BM25 ranking component was implemented directly using the probabilistic ranking formulation proposed by Robertson and Walker [34]. The implementation includes inverse document frequency calculation, document length normalization, and configurable ranking parameters. This approach ensured that all retrieval computations remained transparent and could be adapted to the specific characteristics of OGD metadata.

Similarly, the fuzzy inference engine was implemented as a custom Mamdani-style system supporting multiple membership function types, configurable fuzzy operators, rule evaluation mechanisms, aggregation procedures, and centroid-based defuzzification. Implementing these components directly enabled detailed experimentation with different fuzzy configurations and facilitated the generation of explainable ranking decisions.

The custom implementation strategy contributes to the transparency and interpretability objectives of the research by providing complete control over the ranking process and eliminating dependencies on external black-box libraries.

5.3 Dataset Acquisition Implementation

The dataset acquisition component was responsible for collecting metadata from the Swiss OGD portal. The implementation utilized the CKAN API provided by the *opendata.swiss* platform to access publicly available dataset metadata [10, 9].

The acquisition process retrieved essential metadata attributes including dataset titles, descriptions, tags, groups, organizational information, publication dates, and update timestamps. These metadata elements formed the foundation of the retrieval and ranking framework developed in this research. The selection of metadata attributes was guided by previous research highlighting the importance of metadata quality and completeness for dataset discoverability and retrieval effectiveness [20, 6].

To improve robustness, the acquisition module incorporated validation procedures to handle incomplete or inconsistent metadata records. Retrieved datasets were subsequently transformed into a structured format suitable for metadata processing, indexing, and retrieval operations. Such preprocessing is particularly important in OGD environments where metadata quality may vary across publishers [28, 1].

The acquisition layer was designed to support periodic metadata updates, ensuring that newly published datasets and metadata modifications could be incorporated into the retrieval framework without requiring major system modifications. This capability helps maintain dataset freshness and improves the long-term usability of the retrieval system.

Evaluation Corpus. Although the *opendata.swiss* portal contains substantially more datasets, the experimental evaluation presented in this thesis was conducted using a curated corpus of 500 datasets retrieved through the CKAN API. The corpus consisted primarily of datasets published by the Swiss Federal Statistical Office (FSO), the Canton of Thurgau, and the Swiss National Library. The subset was selected to provide a practical dataset for prototype development and evaluation while maintaining realistic metadata characteristics. Therefore, the reported findings should be interpreted within the scope of this evaluation corpus and should not be generalized to the entire *opendata.swiss* repository without further large-scale validation.

5.4 Metadata Weighting Strategy

The retrieval framework employs a metadata weighting strategy to emphasize the relative importance of different metadata attributes during document construction. Rather than treating all metadata fields equally, the implementation assigns greater importance to fields that are expected to contribute more significantly to dataset discoverability and relevance estimation. Previous IR research has demonstrated that the effectiveness of retrieval models can be improved by appropriately weighting informative document attributes [35, 34].

Dataset titles are considered the most informative metadata element and therefore receive the highest weighting. Tags and groups are also assigned increased importance because they provide structured thematic information that can improve retrieval effectiveness. Descriptions are included to provide contextual information regarding dataset

content, while organizational information is used only as a fallback when structured metadata attributes are unavailable.

The implemented weighting strategy is summarized as follows:

- Title: weight 3
- Tags: weight 2
- Groups: weight 2
- Description: weight 1
- Organization: fallback attribute

During document construction, title information is repeated three times, tags and groups are repeated twice, and descriptions are included once. This weighted representation increases the influence of highly informative metadata attributes while preserving all available metadata content.

The weighting strategy was specifically designed to improve thematic matching between user queries and dataset metadata. By emphasizing metadata fields that are most likely to reflect dataset topics and user intent, the framework supports more effective dataset discovery within the Swiss OGD environment. This design is consistent with established IR principles that prioritize informative content when estimating document relevance [34, 35].

5.5 Custom BM25 Implementation

The retrieval component was implemented using a custom BM25 algorithm specifically developed for the proposed HCIR framework. Although existing BM25 libraries are available, a custom implementation was selected to provide complete control over metadata processing, term weighting, similarity estimation, and integration with the fuzzy ranking framework. This approach also ensured transparency and allowed the retrieval process to be adapted to the specific characteristics of OGD metadata [34].

The implementation follows the probabilistic relevance framework proposed by Robertson and Zaragoza [34]. BM25 was selected because it provides effective handling of term frequency, document length normalization, and relevance estimation while maintaining interpretability and computational efficiency. Compared with traditional term-weighting approaches, BM25 offers a more robust mechanism for ranking documents and has consistently demonstrated strong retrieval performance across IR applications [34, 35].

The inverse document frequency (IDF) used within the implementation is calculated as:

$$IDF = \log \left(\frac{N - df + 0.5}{df + 0.5} + 1 \right) \quad (5.1)$$

where (N) represents the total number of indexed datasets and (df) denotes the document frequency of a query term. This formulation reduces the influence of highly frequent terms while assigning greater importance to terms that are more discriminative within the dataset collection [34].

The implementation employs the standard BM25 parameters $k_1 = 1.5$ and $b = 0.75$, where k_1 represents the term frequency saturation parameter and b represents the document length normalization parameter. The parameter k_1 controls the extent to which repeated occurrences of a query term contribute to the relevance score, while b regulates the influence of document length normalization. These parameter values are widely adopted within IR research and provide a balanced trade-off between retrieval effectiveness and computational efficiency [34].

Unlike traditional document retrieval systems that primarily operate on unstructured text, the proposed implementation applies BM25 to weighted metadata representations constructed from dataset titles, tags, groups, descriptions, and organizational information. This adaptation enables BM25 to operate effectively within the OGD environment while preserving compatibility with the subsequent fuzzy ranking process.

The custom implementation further facilitates integration with metadata completeness assessment, dataset recency evaluation, explainability generation, and fuzzy inference mechanisms. Consequently, BM25 serves as the foundation for thematic similarity estimation within the overall retrieval framework and provides the primary input to the fuzzy ranking component developed in this research.

5.6 Similarity Normalization

The raw BM25 scores generated by the retrieval component are not directly supplied to the fuzzy inference engine. Instead, a normalization step is performed to transform the scores into a bounded similarity measure suitable for fuzzy processing.

The normalization process combines the BM25 score with query coverage information. Query coverage represents the proportion of query terms successfully matched within the metadata representation of a dataset. By incorporating coverage information, the framework rewards datasets that match a larger portion of the user’s query while maintaining the relevance estimation provided by BM25.

The normalized similarity value is calculated as:

$$Similarity = \frac{Score}{Score + 1.0} \times Coverage \quad (5.2)$$

where (Score) represents the raw BM25 retrieval score and (Coverage) represents the proportion of matched query terms. This transformation constrains similarity values to the interval $([0,1])$, making them suitable inputs for the fuzzy inference system.

The normalization process also reduces the influence of extremely large BM25 values that may occur for highly relevant datasets. Consequently, the fuzzy ranking component receives consistent and interpretable similarity values that can be combined with metadata completeness and dataset recency indicators.

The resulting normalized similarity score serves as the thematic similarity input of the fuzzy ranking framework. This design enables the integration of traditional IR techniques with fuzzy logic-based relevance estimation while maintaining transparency and interpretability throughout the ranking process [34, 42, 4].

5.7 Custom Mamdani Fuzzy Inference Engine

The fuzzy ranking component was implemented using a custom Mamdani fuzzy inference engine specifically developed for the requirements of the proposed HCIR framework. Although several fuzzy logic libraries are available, a custom implementation was selected to provide full control over membership functions, rule evaluation, inference procedures, and explainability generation.

The implementation follows the Mamdani fuzzy inference approach, one of the most widely adopted fuzzy reasoning models due to its interpretability and ability to represent human decision-making processes through linguistic rules [24, 42]. This characteristic aligns closely with the objectives of HCIR, where transparency and user understanding are essential.

The fuzzy inference engine consists of four primary stages:

1. Fuzzification
2. Rule Evaluation
3. Aggregation
4. Defuzzification

During fuzzification, numerical inputs are transformed into linguistic variables using predefined membership functions. Rule evaluation applies the fuzzy rule base to determine the degree of activation for each rule. Aggregation combines the outputs of all activated rules, while defuzzification converts the aggregated fuzzy output into a single numerical relevance score.

The custom implementation supports multiple membership function types, including:

- Triangular
- Trapezoidal
- Gaussian
- Sigmoid
- Generalized Bell

Supporting multiple membership functions provides flexibility during system design and experimentation while allowing the fuzzy ranking framework to adapt to different relevance estimation requirements.

The final output generated by the Mamdani inference engine represents the overall relevance score of a dataset. The inference process integrates four complementary relevance indicators: thematic similarity, metadata completeness, dataset recency, and resource availability. Combining these indicators through fuzzy reasoning enables the framework to model gradual relevance relationships while simultaneously considering both topical relevance and important metadata characteristics. The resulting relevance score forms the basis for the final dataset ranking and supports transparent, human-interpretable decision-making, which is consistent with the principles of HCIR and explainable decision support systems [4, 42, 3, 27].

The use of a custom Mamdani inference engine also improves transparency because every ranking decision can be traced back to specific membership functions and fuzzy rules. This capability directly supports the explainability objectives of the proposed HCIR framework and contributes to greater user trust in ranking outcomes [3, 27].

Justification for Selecting the Mamdani Fuzzy Inference System

The selection of the Mamdani fuzzy inference system was guided primarily by the objective of supporting HCIR through transparent and interpretable ranking decisions. Unlike Takagi–Sugeno (TSK) fuzzy systems, which generate mathematical output functions, Mamdani inference represents both inputs and outputs using linguistic variables such as *Low*, *Medium*, and *High*. This linguistic representation enables ranking decisions to be communicated in a manner that is understandable to end users.

Since one of the primary objectives of this research is to improve transparency and explainability, interpretability was considered more important than computational efficiency. The linguistic outputs generated by the Mamdani approach integrate naturally with the explanation component of the proposed framework, allowing users to understand how metadata completeness, keyword relevance, publication recency, and organizational quality jointly influence dataset ranking.

Although Takagi–Sugeno fuzzy systems are widely used in control and optimization applications because of their computational efficiency, the Mamdani approach is generally considered more suitable for decision-support systems in which human interpretation of inference results is required. Consequently, the Mamdani fuzzy inference system was selected as the most appropriate reasoning mechanism for the proposed HCIR framework [42, 38, 12].

Why Learning-to-Rank Was Not Adopted

Learning-to-Rank (LTR) approaches represent an important class of IR algorithms that learn ranking functions directly from labelled training data. Although these techniques have demonstrated strong retrieval performance in many large-scale retrieval tasks, they were not adopted in this research for several reasons.

First, the benchmark developed for this study contains a relatively small number of representative queries and manually reviewed relevance judgments, which is insufficient for training and validating a supervised Learning-to-Rank model. Second, many Learning-to-Rank algorithms operate as complex machine learning models whose ranking decisions are difficult to interpret. Since the objective of this research is to support HCIR through transparent and explainable ranking, interpretability was considered a primary design requirement.

Consequently, a rule-based Mamdani fuzzy inference system was selected because it enables explicit modelling of multiple metadata-based relevance criteria while preserving transparency and allowing users to understand how ranking decisions are generated [42, 8].

Design rationale. The proposed fuzzy inference system combines data-calibrated membership functions with an expert-defined rule base because these two components represent different types of knowledge. Membership functions describe observable characteristics of the metadata, such as similarity scores, metadata completeness, recency, and resource availability. Since these variables are directly measurable from the dataset collection, their breakpoints were calibrated using the statistical distribution of the corpus to ensure that the linguistic categories accurately reflected the characteristics of the available metadata.

In contrast, the fuzzy rules represent human judgement regarding how different relevance factors should be combined during ranking. Such preferences cannot be derived reliably from the relatively small benchmark query set used in this study and therefore require expert knowledge. Defining the rules manually also preserves transparency because every ranking decision can be traced back to explicit IF–THEN rules that are understandable to both developers and users. This combination of data-driven calibration and expert knowledge provides a balance between empirical adaptation and interpretability, consistent with the objectives of HCIR.

5.7.1 Fuzzy Ranking Inputs

The proposed fuzzy ranking framework utilizes four complementary relevance indicators to evaluate dataset quality and relevance. These indicators were selected to address limitations commonly observed in traditional retrieval systems that rely exclusively on textual similarity measures. By incorporating multiple relevance factors, the framework provides a more comprehensive and human-centered assessment of dataset usefulness [3, 4].

The first input is **Thematic Similarity**, which represents the degree of correspondence between the user’s query and the metadata content of a dataset. Thematic similarity is generated by the BM25 retrieval component and subsequently normalized before being supplied to the fuzzy inference engine. This indicator captures the topical relevance of datasets with respect to user information needs [34].

The second input is **Metadata Completeness**. This indicator evaluates the extent to which important metadata attributes are available for a dataset. Datasets containing complete and well-structured metadata generally provide greater contextual information and are easier for users to interpret and assess. Previous studies have identified metadata quality as a critical factor influencing dataset discoverability and reuse within OGD environments [20, 6, 28].

The third input is **Dataset Recency**. Recency measures the freshness of a dataset based on available publication and update information. In many OGD scenarios, users prefer datasets that have been recently updated because they are more likely to reflect current conditions and provide reliable information. Incorporating recency therefore enables the ranking framework to account for temporal relevance in addition to thematic relevance [15, 1].

The fourth input is **Resource Availability**. This indicator evaluates the practical accessibility and usability of a dataset by considering the availability of associated resources, such as downloadable files or supporting data services. Even when a dataset is topically relevant, limited resource availability may reduce its usefulness for users wishing to inspect or reuse the data. Incorporating this criterion therefore extends the ranking process beyond topical relevance by accounting for the practical utility of search results, an important consideration in HCIR and OGD environments [1, 3].

Before fuzzy inference is performed, each input variable is transformed into fuzzy linguistic variables through the fuzzification process. This transformation allows quantitative measurements to be represented using human-interpretable concepts such as *Low*, *Medium*, and *High*. The resulting fuzzy representations provide the foundation for rule-based reasoning and relevance estimation within the fuzzy ranking framework [42, 24].

By combining thematic similarity, metadata completeness, dataset recency, and resource availability, the proposed framework evaluates dataset relevance from multiple

complementary perspectives rather than relying solely on textual matching. This multi-criteria approach better reflects how users assess dataset usefulness during exploratory search and supports the objectives of HCIR by integrating relevance, metadata quality, temporal validity, and practical accessibility into a single interpretable ranking process [3, 26, 4].

5.7.2 Explanation Generation

A key objective of the proposed HCIR framework is to improve transparency and user understanding during the dataset discovery process. Traditional retrieval systems typically present ranked results without explaining the factors that influenced ranking decisions. As a result, users may find it difficult to understand why particular datasets appear at the top of search results. Explainability addresses this limitation by providing meaningful descriptions of the reasoning process underlying relevance estimation [27, 3].

The explanation generation component was designed to transform fuzzy reasoning outcomes into human-readable descriptions. Rather than presenting only numerical relevance scores, the system identifies the primary factors contributing to dataset rankings and communicates these factors through concise textual explanations.

Explanations are generated directly from the outputs of the ranking process rather than from predefined textual templates. For each ranked dataset, the system identifies the relevance indicators that contributed most strongly to the final score, including thematic similarity, metadata completeness, dataset recency, and resource availability. These values are combined with information derived from the fuzzy inference process to produce faithful explanations that accurately reflect the factors influencing the ranking decision. Consequently, the explanation component communicates the actual reasoning underlying the computed relevance score rather than presenting a generic post-hoc justification, thereby improving transparency and user trust [27, 8, 3].

The explanation generation process follows three primary steps:

1. Identification of dominant ranking factors.
2. Analysis of fuzzy inference outcomes.
3. Generation of human-readable explanations.

This approach enables users to understand how different relevance indicators contributed to ranking decisions. By exposing the reasoning process, the system promotes transparency and supports informed decision-making during exploratory search activities.

The explanation module is tightly integrated with the ranking pipeline rather than operating as an independent post-processing component. After the fuzzy inference engine

computes the final relevance score, the system records the contribution of each ranking factor, including thematic similarity, metadata completeness, dataset recency, and resource availability. These recorded values are subsequently transformed into human-readable explanations presented alongside the ranked results. Because the explanations are derived directly from the same metadata features and fuzzy reasoning process used for ranking, they faithfully represent the underlying decision-making process rather than providing approximate or post-hoc justifications.

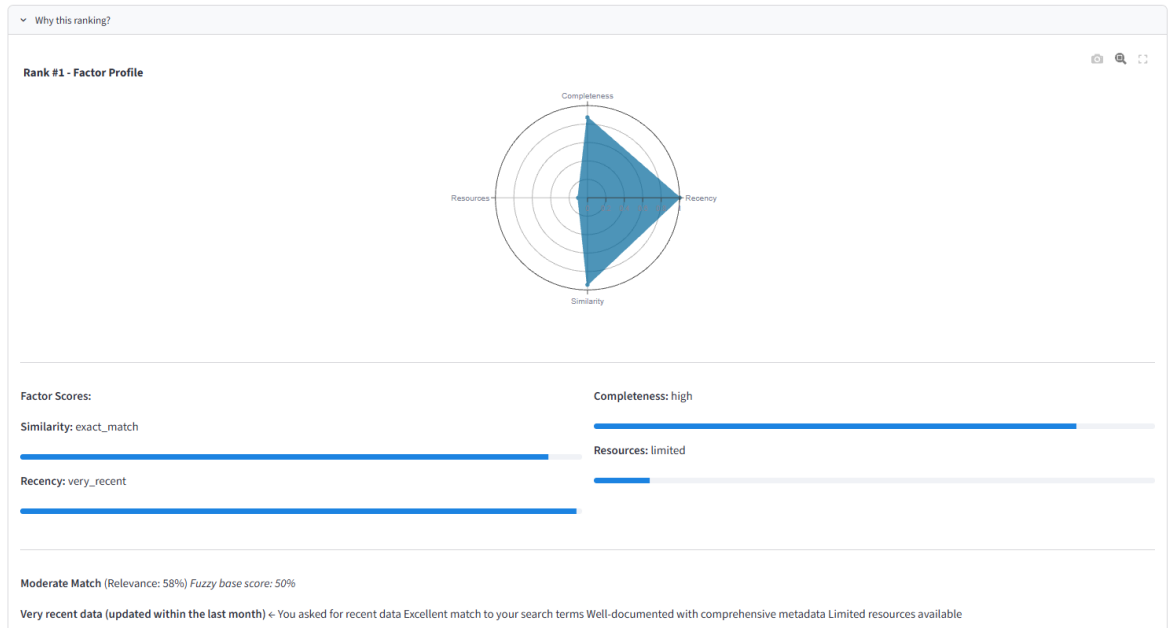


Figure 5.1: Explanation interface illustrating the factors contributing to dataset relevance, including thematic similarity, metadata completeness, recency, and resource availability.

Figure 5.1 illustrates the explanation interface developed to improve transparency within the proposed HCIR framework. For each ranked dataset, the system presents a detailed breakdown of the factors contributing to the final relevance score, including thematic similarity, metadata completeness, dataset recency, and resource availability. A visual radar chart summarizes the relative contribution of each factor, while textual explanations provide additional context regarding the ranking decision. By exposing the reasoning process behind relevance estimation, the explanation component enables users to better understand why particular datasets appear at specific ranking positions. This functionality supports transparency, interpretability, and user trust, which are central objectives of HCIR and explainable information systems [3, 27, 42].

The use of fuzzy logic further enhances explainability because the underlying reasoning process is based on linguistic variables and interpretable rule structures. Unlike many complex machine learning models that operate as black boxes, fuzzy inference systems provide clear relationships between inputs, rules, and outputs [42, 4]. Consequently,

users can better understand how relevance scores are produced and how different dataset characteristics influence ranking outcomes.

From a HCIR perspective, explainability contributes to user trust, system transparency, and improved interaction quality. Providing explanations alongside search results enables users to evaluate dataset relevance more effectively and supports the broader objective of making OGD more accessible and understandable [3, 27].

5.8 User Interface Implementation

The UI was implemented using Streamlit to provide an interactive and user-friendly environment for dataset discovery and exploration. Streamlit was selected because it supports rapid development of web-based applications while enabling seamless integration with Python-based retrieval and ranking components.

The interface was designed according to the principles of HCIR, emphasizing usability, transparency, and user understanding throughout the search process [3, 26]. Users can submit natural-language queries, inspect ranked datasets, and access explanations describing the factors that influenced ranking decisions.

The search interface consists of several interconnected components. The query input component allows users to enter information needs using natural language expressions. Submitted queries are processed by the retrieval framework, which generates thematic similarity estimates and subsequently applies fuzzy relevance estimation.

Search results are presented in ranked order according to the relevance scores generated by the fuzzy inference engine. For each dataset, the interface displays key metadata attributes including titles, descriptions, tags, groups, and organizational information. Presenting metadata directly within the interface enables users to quickly evaluate dataset relevance without requiring additional navigation.

In addition to ranked results, the interface provides explanation panels that communicate the primary factors contributing to relevance estimation. These explanations are generated from the fuzzy reasoning process and help users understand how thematic similarity, metadata completeness, and dataset recency influenced ranking outcomes. Providing such explanations improves transparency and supports informed decision-making during exploratory search activities [27, 3].

The interface also supports exploratory dataset discovery by enabling users to compare datasets, examine metadata characteristics, and refine search strategies. This functionality aligns with HCIR principles that view information seeking as an interactive and evolving process rather than a simple query-response interaction [26].

Overall, the UI serves as the primary interaction point between users and the retrieval framework, integrating retrieval, fuzzy ranking, and explainability components into a unified environment for OGD discovery.

5.9 Integration of Components

The proposed HCIR framework consists of multiple interconnected components that operate together to support dataset discovery within the Swiss OGD environment. The integration process combines metadata acquisition, BM25 retrieval, similarity normalization, fuzzy relevance estimation, explanation generation, and user interaction within a unified architecture.

The retrieval process begins when a user submits a query through the interface. The query is forwarded to the retrieval component, where BM25 is applied to estimate thematic similarity between the query and available dataset metadata. The resulting similarity scores provide an initial assessment of dataset relevance based on textual matching principles [34].

Following retrieval, the similarity normalization component transforms raw BM25 scores into bounded similarity values suitable for fuzzy processing. The normalized thematic similarity score is subsequently combined with metadata completeness and dataset recency indicators to create the input variables required by the fuzzy inference engine.

The fuzzy ranking component evaluates these inputs through fuzzification, rule evaluation, aggregation, and defuzzification. This process produces a final relevance score that reflects multiple dimensions of dataset quality and relevance rather than relying solely on textual similarity [42, 24, 4].

The explanation generation component operates in parallel with the ranking process. Information obtained from the fuzzy inference engine is analyzed to identify the dominant factors contributing to ranking decisions. These factors are subsequently transformed into human-readable explanations that accompany search results and improve transparency [27].

Finally, the UI presents ranked datasets together with relevant metadata and generated explanations. Users can inspect search results, compare datasets, and better understand the reasoning process underlying relevance estimation. This integrated workflow supports the objectives of HCIR by combining retrieval effectiveness, explainability, and user-centered interaction within a single framework [3, 26].

5.10 Testing and Validation

Testing and validation activities were conducted throughout the development process to ensure the reliability, correctness, and usability of the proposed HCIR framework. The objective of testing was to verify that individual components functioned correctly and that the integrated system produced consistent and meaningful retrieval results.

Several levels of testing were performed, including functional testing, integration testing, retrieval validation, fuzzy inference validation, and UI testing. This multi-level ap-

proach ensured that both individual components and the complete framework operated as intended.

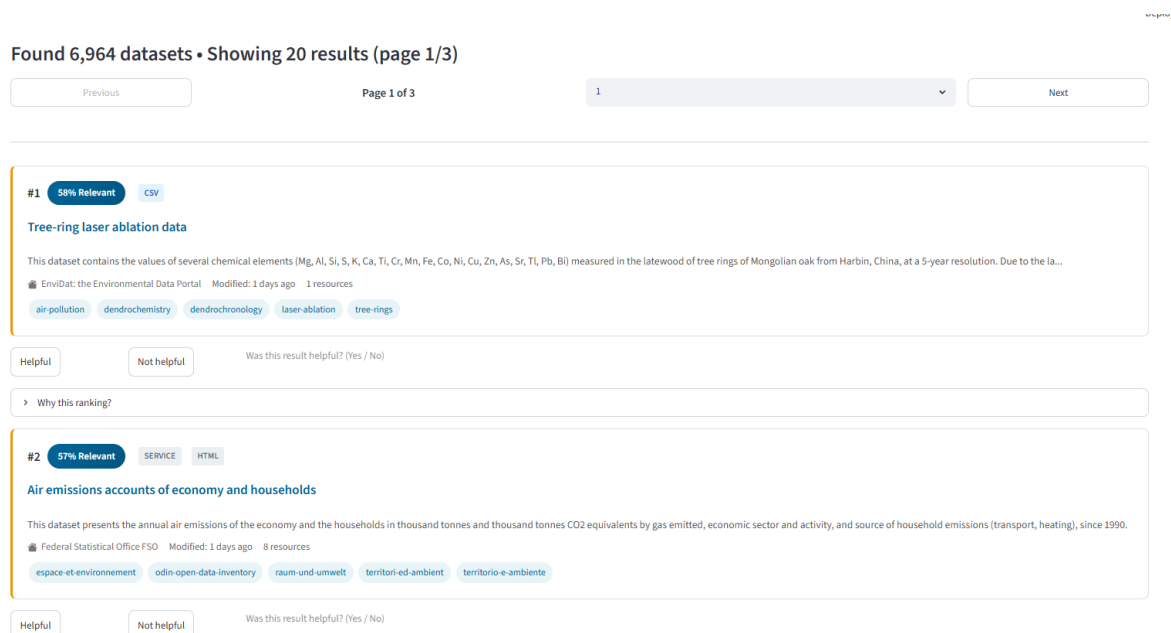


Figure 5.2: Search results interface displaying ranked datasets, relevance scores, metadata attributes, and access to explanation functionality.

Figure 5.2 presents the search results interface of the proposed Fuzzy HCIR system. Retrieved datasets are displayed in ranked order according to the relevance scores generated by the fuzzy inference engine. Each result includes key metadata attributes such as dataset title, description, publisher information, update date, resource count, and thematic tags. The interface also provides relevance indicators and direct access to explanation functionality, enabling users to understand why a dataset was assigned a particular ranking position. By combining ranked retrieval with transparent presentation of metadata and relevance information, the interface supports effective dataset exploration and aligns with HCIR principles that emphasize usability, transparency, and informed decision-making [3, 26, 27].

5.10.1 Functional Testing

Functional testing was conducted to verify the behavior of individual system components. The dataset acquisition module, metadata processing component, BM25 retrieval engine, fuzzy inference engine, and explanation generation module were tested independently to confirm correct operation under various input conditions.

Particular attention was given to validating metadata processing procedures, similarity calculations, fuzzy rule evaluation, and explanation generation mechanisms. Test cases were designed to verify that each component produced expected outputs and handled incomplete or inconsistent metadata appropriately.

5.10.2 Integration Testing

Integration testing was performed to verify communication between system components. The complete workflow, beginning with user query submission and ending with ranked dataset presentation, was evaluated to ensure seamless interaction among retrieval, ranking, explanation, and UI modules.

Integration testing confirmed that similarity values generated by the BM25 retrieval component were correctly transferred to the fuzzy inference engine and that final relevance scores were successfully communicated to the explanation and visualization components.

5.10.3 Retrieval Validation

Retrieval validation focused on assessing whether the BM25 component produced meaningful thematic similarity estimates. A variety of test queries were submitted to the system, and the resulting rankings were manually inspected to determine whether highly relevant datasets appeared near the top of the ranking list.

The validation process demonstrated that the retrieval component effectively identified datasets whose metadata closely matched user information needs. These results provided confidence that thematic similarity estimates could serve as reliable inputs to the fuzzy ranking framework [34].

5.10.4 Fuzzy Inference Validation

Validation of the fuzzy inference engine focused on verifying the correctness of fuzzification, rule evaluation, aggregation, and defuzzification procedures. Test scenarios representing different combinations of thematic similarity, metadata completeness, and dataset recency were evaluated to ensure that the resulting relevance scores were consistent with the intended rule base.

The validation process confirmed that the fuzzy ranking framework produced interpretable relevance estimates and appropriately reflected the influence of multiple ranking factors. These findings are consistent with established principles of fuzzy reasoning and fuzzy IR [42, 24, 4].

5.10.5 User Interface Testing

UI testing was conducted to verify usability, navigation, and interaction functionality. The search interface, ranking display, metadata presentation, and explanation components were evaluated to ensure that users could successfully perform search and exploration tasks.

Testing confirmed that the interface supported efficient query submission, result inspection, and explanation interpretation. The interface therefore provided an appropriate

environment for subsequent user evaluation activities conducted during the assessment phase of the research.

Overall, the testing and validation activities demonstrated that the proposed framework operated reliably and that all major components functioned correctly both individually and as part of the integrated system.

5.11 Implementation Challenges

The development of the proposed HCIR framework involved several technical and methodological challenges. These challenges emerged during metadata acquisition, retrieval model implementation, fuzzy system design, and explainability generation. Addressing these challenges was necessary to ensure the reliability, effectiveness, and usability of the final prototype.

5.11.1 Metadata Quality and Consistency

One of the primary challenges involved the quality and consistency of metadata available through the Swiss OGD portal. Dataset publishers often provide metadata with varying levels of completeness, detail, and standardization. Some datasets contained comprehensive metadata descriptions, while others lacked important attributes such as tags, groups, or detailed descriptions.

This inconsistency complicated both retrieval and ranking processes because metadata quality directly influences dataset discoverability. To mitigate this issue, preprocessing and validation procedures were incorporated into the acquisition pipeline to improve metadata handling and ensure that incomplete records could still be processed effectively [1, 20, 28].

5.11.2 Balancing Multiple Relevance Factors

Traditional retrieval systems typically rely on textual similarity as the primary ranking criterion. In contrast, the proposed framework integrates thematic similarity, metadata completeness, and dataset recency and resource availability. Determining how these factors should interact during relevance estimation represented a significant design challenge.

The fuzzy inference approach was selected to address this issue because it provides a flexible mechanism for combining multiple relevance indicators while preserving interpretability. Designing an appropriate rule base required iterative refinement to ensure that ranking outcomes reflected realistic user expectations and relevance judgments [42, 24, 4].

5.11.3 Design of Membership Functions

Another challenge involved the definition of membership functions used during fuzzification. Membership functions determine how numerical inputs are transformed into linguistic concepts such as *Low*, *Medium*, and *High*. Poorly defined membership functions can lead to unrealistic relevance estimations and unstable ranking behavior.

Several membership function configurations were evaluated during development to identify representations that produced meaningful and interpretable ranking outcomes. The final configuration was selected based on consistency, interpretability, and compatibility with the retrieval objectives of the framework [43, 42].

5.11.4 Explainability and User Understanding

Providing meaningful explanations for ranking decisions presented an additional challenge. While numerical relevance scores can indicate dataset importance, they do not necessarily help users understand why particular datasets were ranked highly.

To address this limitation, the explanation generation component was designed to identify dominant ranking factors and translate them into human-readable descriptions. This approach improves transparency and aligns with HCIR principles that emphasize user understanding and trust [27, 3].

5.11.5 System Integration

Integrating multiple components into a unified retrieval framework also required careful coordination. Metadata acquisition, BM25 retrieval, similarity normalization, fuzzy inference, explanation generation, and UI components needed to communicate efficiently while maintaining consistent data representations.

Ensuring compatibility among these components required iterative testing and refinement throughout the development process. Successful integration was essential for achieving the overall objective of providing an effective and explainable dataset retrieval environment.

Despite these challenges, the final implementation successfully integrated all major components into a functional prototype capable of supporting HCIR for Swiss OGD.

5.12 Chapter Summary

This chapter presented the implementation of the proposed HCIR framework for Swiss OGD. The implementation process included dataset acquisition, metadata processing, metadata weighting, custom BM25 retrieval, similarity normalization, fuzzy relevance estimation, explanation generation, and UI development.

A custom BM25 retrieval component was implemented to estimate thematic similarity between user queries and dataset metadata. The resulting similarity scores were normalized and combined with metadata completeness, dataset recency, and resource availability indicators through a custom Mamdani fuzzy inference engine. This approach enabled the framework to evaluate dataset relevance using multiple criteria while maintaining transparency and interpretability [34, 42, 24, 4].

The chapter further described the implementation of the explanation generation component and the Streamlit-based UI, both of which support the HCIR objectives of transparency, explainability, and user understanding [3, 27].

Finally, testing and validation activities confirmed the correct operation of individual components and the integrated system. The implementation challenges encountered during development were also discussed, together with the strategies adopted to address them.

Having presented the design and implementation of the proposed framework, the next chapter evaluates the system through experimental assessment and user-based evaluation to determine its effectiveness in supporting dataset discovery and retrieval within the Swiss OGD environment.

Chapter 6

Evaluation Methodology

This chapter presents the methodology used to evaluate the proposed HCIR framework. The objective of the evaluation is to determine whether the system effectively improves dataset discoverability while supporting transparency, explainability, and user-centered interaction. The chapter describes the evaluation objectives, participant selection process, data collection methods, evaluation metrics, and analysis procedures used to assess the performance and usability of the proposed retrieval framework.

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6.1 Evaluation Objectives

The evaluation was designed to assess the extent to which the proposed retrieval framework achieves the objectives established in Chapter 1. Specifically, the evaluation sought to determine whether the integration of BM25 retrieval, fuzzy logic-based ranking, and explainability mechanisms contributes to improved dataset discovery and user experience.

The evaluation objectives are summarized as follows:

1. Assess the effectiveness of the retrieval framework in identifying relevant datasets.
2. Evaluate user perceptions of system usability.
3. Measure the perceived transparency of ranking decisions.
4. Assess the usefulness of generated explanations.
5. Investigate the contribution of fuzzy logic to HCIR.

6. Identify strengths and limitations of the proposed framework.

6.2 Evaluation Strategy

The proposed HCIR framework was evaluated using a mixed-method evaluation strategy consisting of two complementary components: (1) a quantitative Information Retrieval (IR) benchmark and (2) a formative user-centred questionnaire. This evaluation strategy is consistent with Design Science Research (DSR), which recommends assessing both the technical performance of an artefact and its perceived usefulness to end users [13, 32].

The quantitative evaluation was conducted using a frozen metadata snapshot of the Swiss Open Government Data (OGD) portal to ensure reproducibility. All evaluated retrieval systems operated on the same metadata corpus, benchmark query set, and relevance judgments under identical experimental conditions. To minimise system-specific assessment bias, a pooled evaluation methodology was adopted. For each benchmark query, the highest-ranked candidate datasets returned by all evaluated retrieval systems were combined into a common candidate pool, and duplicate datasets were removed before relevance assessment. This pooling strategy ensured that the relevance assessment considered datasets retrieved by all systems rather than evaluating each system independently.

Relevance judgments were assigned by the author as a single assessor using a three-level graded relevance scale consisting of 0 (not relevant), 1 (partially relevant), and 2 (highly relevant). Following the initial assessment, the assigned labels were reviewed during a separate consolidation pass before producing the final `ground_truth_final.json` file used throughout the benchmark evaluation. The published judgment files are included with the evaluation artifacts to support transparency, reproducibility, and independent inspection of the assessment process.

The benchmark was executed as a reranking experiment on the frozen metadata snapshot. For every benchmark query, all evaluated systems ranked the same candidate datasets, allowing direct comparison of ranking quality while avoiding changes caused by updates to the live portal. Retrieval effectiveness was measured using Mean Average Precision (MAP), Precision at rank five (P@5), Normalized Discounted Cumulative Gain at rank ten (nDCG@10), and Mean Reciprocal Rank (MRR), providing complementary measures of ranking quality, early precision, graded relevance, and the position of the first relevant result [25, 16].

The second component of the evaluation consisted of a formative user-centred questionnaire involving ten Master’s students. Participants completed representative search tasks using the prototype before responding to a structured questionnaire assessing usability, transparency, explanation clarity, and perceived usefulness. The questionnaire complements the quantitative benchmark by providing descriptive evidence regarding

users' perceptions of the proposed framework rather than measuring causal effects on user performance or decision-making.

This evaluation strategy combines an objective retrieval benchmark with a formative user study to assess both the technical behaviour of the proposed HCIR framework and users' perceptions of its transparency and usability. Because the relevance judgments were assigned by a single assessor, inter-rater agreement measures such as Cohen's kappa are not applicable. To support transparency, the complete benchmark configuration, final judgment file, and evaluation artifacts are published with the project, allowing the evaluation procedure to be independently reviewed and reproduced.

6.3 Participants

The evaluation involved participants with varying levels of familiarity with IR systems, OGD, and data analysis. Including participants from diverse backgrounds was important because OGD portals are intended for a broad range of users, including researchers, students, policymakers, journalists, and citizens.

The participant group was intentionally selected because Master's students and researchers regularly perform exploratory IR tasks and possess sufficient technical literacy to critically evaluate retrieval quality, explanation mechanisms, and system transparency. As the objective of this study was to conduct a formative evaluation of a research prototype rather than a large-scale usability validation, this participant group provided appropriate feedback on the proposed HCIR framework. Nevertheless, the findings should not be interpreted as representative of all OGD users, and future studies should include a broader range of stakeholders, including public administrators, journalists, policymakers, and citizens.

The participant group was selected using purposive sampling because the study aimed to obtain informed feedback from individuals capable of critically evaluating IR systems and OGD search tasks. Master's students in Computer Science were considered appropriate participants because they possess sufficient technical knowledge to understand retrieval behaviour while still representing prospective end users of data discovery platforms.

The purpose of the user study was formative rather than summative. Consequently, the results are intended to identify strengths, weaknesses, and opportunities for improving the proposed HCIR framework rather than to make statistically generalizable claims about the wider population of OGD users. This interpretation is consistent with established HCI evaluation practices, where small participant groups are commonly used to evaluate usability and support iterative refinement of interactive systems [? 5].

6.4 Participant Demographics

Participants completed a series of representative search tasks designed to reflect realistic information needs encountered when exploring Swiss OGD. The tasks were selected to cover both targeted retrieval scenarios, in which users searched for specific datasets, and exploratory search scenarios, in which users refined broad or partially specified information needs through interaction with the retrieval system. This combination enabled the evaluation to assess not only retrieval effectiveness but also the usability, transparency, and explainability of the proposed HCIR framework under realistic search conditions [26, 40, 3].

The collected demographic information included:

- Age group
- Gender
- Educational background
- Professional background
- Experience with OGD
- Experience with search and retrieval systems

The demographic analysis provides insight into the diversity of participants and supports the interpretation of evaluation findings presented in Chapter 7.

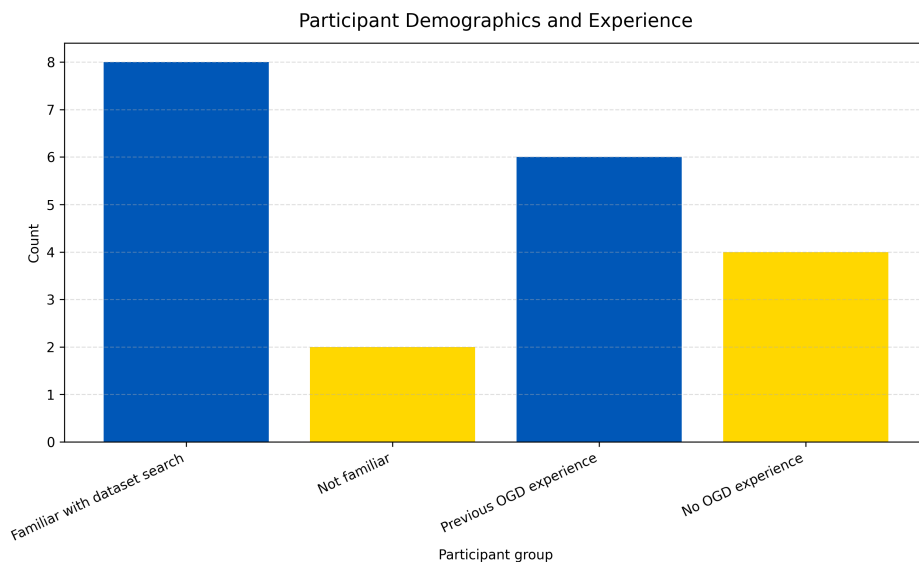


Figure 6.1: Participant demographics

Figure 6.1 presents the demographic characteristics of the 10 participants who took part in the evaluation study. The participant group included individuals with different

educational backgrounds, levels of technical expertise, and varying familiarity with IR systems. This diversity was considered important because OGD portals are intended to support a broad range of users with different information needs and levels of experience.

Although the evaluation involved a relatively small sample size, the inclusion of participants with diverse backgrounds provided valuable insights into the usability, transparency, and explainability of the proposed retrieval framework. The demographic distribution therefore supports the assessment of the system from a HCIR perspective, where the needs and perceptions of different user groups are particularly important.

The evaluation involved 10 participants, the majority of whom were students and researchers with varying levels of experience in IR and OGD. The demographic distribution indicates that participants represented different age groups and educational backgrounds, allowing the proposed framework to be evaluated from multiple perspectives. Such diversity is beneficial because it reflects the heterogeneous user population that may interact with OGD portals in practice.

The evaluation tasks were designed to encourage participants to interact naturally with the retrieval system while assessing multiple aspects of the search experience. During task completion, participants considered the relevance of retrieved datasets, the usefulness of the generated explanations, the clarity of metadata presentation, the ease of interaction, and their overall confidence in the ranking decisions. These observations complemented the quantitative retrieval evaluation by providing formative evidence regarding the usability and interpretability of the proposed framework [13, 32, 27].

The demographic characteristics indicate that the participant group possessed sufficient academic and technical background to complete the evaluation tasks and provide informed feedback on the retrieval interface. Although the participants cannot be considered representative of all potential OGD users, they constitute an appropriate sample for a formative usability evaluation because they were capable of understanding dataset search tasks, interpreting retrieval results, and evaluating explanation quality.

The relatively homogeneous participant group also reduced variability caused by differences in technical expertise, allowing the evaluation to focus on the behaviour of the proposed retrieval framework rather than differences in user experience. Consequently, the findings should be interpreted as evidence supporting the usability and transparency of the proposed system rather than as statistically generalisable conclusions for the broader population of OGD users.

6.5 Evaluation Procedure

The evaluation followed a reproducible multi-stage procedure designed to ensure a fair and consistent comparison between all evaluated retrieval systems. The complete benchmark was executed using a frozen metadata snapshot of the Swiss Open Government Data

(OGD) portal, allowing every retrieval approach to operate on the same evaluation corpus under identical experimental conditions. Using a common corpus eliminated variability caused by updates to the live portal and ensured that the reported benchmark results could be reproduced.

The retrieval benchmark was implemented as a reranking experiment. For each benchmark query, candidate datasets returned by the evaluated retrieval systems were pooled into a common candidate set, and duplicate datasets were removed. This pooling procedure ensured that the relevance assessment considered datasets retrieved across all configured systems while providing a common evaluation basis for every ranking method.

Relevance judgments were assigned by the author as a single assessor using a three-level graded relevance scale consisting of 0 (not relevant), 1 (partially relevant), and 2 (highly relevant). After the initial assessment, the assigned grades were reviewed during a separate consolidation pass before generating the final `ground_truth_final.json` file used throughout the quantitative evaluation. This final judgment file served as the authoritative source for all retrieval metrics reported in Chapter 7. The complete judgment files are published with the evaluation artifacts to support transparency and allow independent inspection of the assessment process.

Following construction of the final ground truth, every retrieval system was executed using the same benchmark query set and identical relevance judgments. Retrieval effectiveness was evaluated using Mean Average Precision (MAP), Precision at rank five (P@5), Normalized Discounted Cumulative Gain at rank ten (nDCG@10), and Mean Reciprocal Rank (MRR), providing complementary measures of ranking quality, early precision, graded relevance, and first relevant result position [25, 16].

To compare retrieval performance across systems, paired statistical analyses were performed using the Wilcoxon signed-rank test with Holm–Bonferroni correction for multiple comparisons. Confidence intervals, per-query analyses, and sensitivity analyses were additionally conducted to examine the robustness of the observed retrieval behaviour under the evaluated experimental configuration.

The robustness of the fuzzy ranking framework was further examined through sensitivity analysis using alternative rule-weight configurations together with controlled perturbations of the membership-function parameters. In addition, an ablation experiment compared the complete fuzzy inference framework with an equivalent linear weighted ranking model while keeping the underlying feature extraction process unchanged. These experiments were designed to evaluate the stability of the ranking behaviour and to isolate the contribution of the fuzzy inference mechanism.

The user-centred evaluation complemented the retrieval benchmark. After completing representative search tasks with the prototype, participants completed a structured questionnaire measuring perceived usability, transparency, explanation clarity, and overall usefulness of the system. The questionnaire results provide descriptive evidence regard-

ing participants’ perceptions of the prototype and complement the quantitative retrieval benchmark without establishing causal relationships.

Because the relevance judgments were produced by a single assessor, inter-rater agreement measures such as Cohen’s kappa and percentage agreement were not applicable. This represents a limitation of the evaluation methodology and is discussed further in Chapter 8. Publishing the benchmark configuration, judgment files, and evaluation outputs provides an audit trail that allows the complete evaluation process to be independently examined and reproduced.

6.6 Evaluation Questionnaire

Following completion of the evaluation tasks, participants completed a structured questionnaire designed to assess their perceptions of the proposed HCIR framework. The questionnaire complemented the quantitative retrieval evaluation by capturing user perceptions of usability, transparency, explainability, trust, and overall satisfaction. These dimensions were selected because the primary objective of the proposed framework extends beyond retrieval effectiveness to include the development of an interpretable and user-centered search experience [3, 27].

The questionnaire included statements such as:

- The system was easy to use.
- The retrieved datasets were relevant to my search query.
- The explanations helped me understand ranking decisions.
- The system appeared transparent and trustworthy.
- The retrieval results were useful for my information needs.

The questionnaire responses were analysed descriptively by computing summary statistics for each evaluation dimension. Open-ended responses were additionally reviewed to identify recurring observations regarding the strengths and limitations of the proposed framework. These qualitative findings were used to complement the quantitative retrieval results rather than to establish statistical generalisations, consistent with the formative evaluation approach adopted in this study [13, 32].

The questionnaire was designed to complement the quantitative IR evaluation by capturing aspects of the user experience that cannot be measured using retrieval effectiveness metrics alone. While objective metrics such as Mean Average Precision (MAP), Precision@5, Mean Reciprocal Rank (MRR), and nDCG@10 quantify ranking performance, the questionnaire assessed participants’ perceptions of usability, transparency,

explanation quality, and overall satisfaction when interacting with the proposed retrieval framework. Combining objective retrieval evaluation with subjective user feedback provides a more comprehensive assessment of HCIR systems by considering both algorithmic performance and user experience [11, 6].

6.7 Evaluation Metrics

Retrieval effectiveness was evaluated using multiple complementary IR metrics to capture different aspects of ranking quality. Rather than relying on a single performance measure, the evaluation employed Mean Average Precision (MAP), Precision at five (P@5), Normalized Discounted Cumulative Gain at rank ten (nDCG@10), and Mean Reciprocal Rank (MRR). Together, these metrics provide a comprehensive assessment of retrieval performance from both system-oriented and user-oriented perspectives [25, 16].

Mean Average Precision (MAP) measures the overall quality of the ranked result list by averaging the precision obtained whenever a relevant document is retrieved. MAP rewards retrieval systems that consistently rank relevant datasets near the top of the result list and is widely regarded as one of the standard evaluation measures in IR research [25].

Precision at five (P@5) evaluates the proportion of relevant datasets among the first five retrieved results. Since users typically examine only the highest-ranked search results, P@5 provides an estimate of the practical usefulness of the ranking for real-world search scenarios [2].

Normalized Discounted Cumulative Gain at rank ten (nDCG@10) evaluates ranking quality while accounting for graded relevance judgments and the position of relevant datasets within the ranked list. Highly relevant datasets appearing near the top contribute more strongly to the final score than equally relevant datasets appearing lower in the ranking, making nDCG particularly appropriate for graded relevance evaluation [16].

Mean Reciprocal Rank (MRR) measures how quickly the first relevant dataset appears in the ranked results by computing the reciprocal of its rank position. This metric reflects how efficiently users can locate useful information during exploratory dataset search [39].

To evaluate whether observed performance differences between retrieval systems were statistically meaningful, paired Wilcoxon signed-rank tests were performed on the per-query evaluation results. Holm–Bonferroni correction was subsequently applied to adjust for multiple pairwise comparisons and to control the family-wise error rate [41, 14]. In addition, 95% bootstrap confidence intervals were estimated using query-level resampling, and win/loss analysis was conducted to compare retrieval systems on individual benchmark queries. Together, these complementary analyses provide stronger evidence than aggregate metric values alone by quantifying both statistical significance and the robustness of the observed performance differences.

Although the retrieval metrics quantify different aspects of ranking quality, they do not by themselves indicate whether observed differences between retrieval systems are statistically meaningful. Therefore, the metric values reported in Chapter 7 are complemented by non-parametric statistical analyses, including the Wilcoxon signed-rank test, Holm–Bonferroni correction for multiple comparisons, and bootstrap confidence intervals. These analyses provide additional evidence regarding the reliability and robustness of the observed retrieval performance differences rather than introducing new evaluation metrics.

6.8 Data Analysis

The quantitative retrieval results were analysed using both descriptive and inferential statistical techniques to provide a comprehensive assessment of retrieval performance. Aggregate evaluation metrics were computed for each retrieval system across the complete benchmark query set, including Mean Average Precision (MAP), Precision at rank five (P@5), Normalized Discounted Cumulative Gain at rank ten (nDCG@10), and Mean Reciprocal Rank (MRR). These complementary metrics evaluate overall ranking quality, early precision, graded relevance, and the position of the first relevant result within the ranked list [25, 16].

Because every retrieval system was evaluated using the same benchmark queries, paired statistical comparisons were performed using the Wilcoxon signed-rank test. This non-parametric test is appropriate for comparing paired retrieval results without assuming normally distributed performance differences [41]. To control the family-wise error rate arising from multiple pairwise comparisons, Holm–Bonferroni correction was applied to all reported significance tests [14].

Confidence intervals were computed for the reported retrieval metrics to quantify uncertainty in the estimated performance values. In addition, per-query analyses and win–loss comparisons were performed to identify query-specific differences between retrieval systems and to provide a more detailed interpretation of aggregate benchmark results.

The robustness of the proposed fuzzy ranking framework was further evaluated through sensitivity analysis. Alternative rule-weight configurations, controlled perturbations of the membership-function breakpoints, and an ablation experiment were conducted to determine whether the observed retrieval behaviour remained stable under reasonable modifications to the fuzzy inference configuration. These analyses complement the primary benchmark by evaluating the robustness of the implemented ranking framework rather than introducing additional retrieval metrics.

Questionnaire responses from the user study were analysed descriptively using summary statistics, including mean scores and standard deviations for each questionnaire

item. These descriptive results were used to characterise participants’ perceptions of usability, transparency, explanation clarity, and overall usefulness of the proposed HCIR framework. Because the user study employed a single-condition formative design, the questionnaire findings are interpreted as descriptive evidence of participant perceptions rather than causal evidence of improvements in user behaviour or decision-making.

The relevance judgments used for the retrieval benchmark were assigned by the author as a single assessor and subsequently reviewed during a separate consolidation pass before generating the final ground-truth file. Consequently, inter-rater agreement measures such as Cohen’s kappa and percentage agreement were not applicable. Instead, transparency and reproducibility are supported through publication of the benchmark configuration, relevance judgment files, and evaluation artifacts, enabling independent inspection of the complete evaluation process.

6.9 Reliability and Validity

Several measures were adopted to strengthen the reliability and validity of the evaluation conducted in this study. For the quantitative retrieval benchmark, all evaluated retrieval systems operated on the same frozen metadata snapshot, benchmark query set, and final relevance judgment file under identical experimental conditions. Using a common evaluation corpus and fixed benchmark configuration ensured that observed differences between systems resulted from their ranking strategies rather than changes in the underlying data or evaluation procedure.

The benchmark was implemented as a reranking experiment in which all systems evaluated the same pooled candidate datasets. Retrieval effectiveness was measured using established Information Retrieval metrics, including Mean Average Precision (MAP), Precision at rank five (P@5), Normalized Discounted Cumulative Gain at rank ten (nDCG@10), and Mean Reciprocal Rank (MRR), providing complementary perspectives on ranking quality and retrieval effectiveness [25, 16]. Statistical analyses using the Wilcoxon signed-rank test together with Holm–Bonferroni correction and confidence intervals further supported interpretation of the observed differences between retrieval systems.

Relevance judgments were assigned by the author as a single assessor and subsequently reviewed during a separate consolidation pass before generating the final `ground_truth_final.json` file used throughout the benchmark. Because only one assessor performed the relevance assessment, inter-rater agreement statistics were not applicable. This represents a potential source of assessment bias and should be considered when interpreting the reported retrieval results. To improve transparency and reproducibility, the benchmark configuration, relevance judgment files, evaluation scripts, and generated outputs are published with the project, enabling independent inspection and reproduction of the evaluation

process.

The formative user evaluation complemented the quantitative benchmark by examining participants' perceptions of usability, transparency, explanation clarity, and overall usefulness. All participants completed the same evaluation tasks and questionnaire under consistent conditions. As the participant group consisted of ten Master's students with broadly similar technical backgrounds, the findings provide formative evidence regarding user perceptions of the prototype but should not be interpreted as statistically representative of the broader population of Open Government Data users.

Construct validity was supported by selecting evaluation measures that correspond directly to the research objectives. Standard Information Retrieval metrics were used to evaluate retrieval effectiveness, while the questionnaire assessed perceived usability, transparency, explanation quality, and overall usefulness. Together, these complementary forms of evaluation provide evidence regarding both the technical behaviour of the proposed HCIR framework and users' perceptions of the system while remaining consistent with the Design Science Research methodology adopted in this study.

6.10 Ethical Considerations

Ethical considerations were taken into account throughout the evaluation process. Participation was voluntary, and participants were informed about the purpose of the study before data collection began.

No personally identifiable information was collected beyond demographic characteristics necessary for analysis. Participant responses were anonymized and used solely for academic research purposes.

The evaluation adhered to principles of confidentiality, informed consent, and responsible research conduct.

6.11 Chapter Summary

This chapter presented the methodology used to evaluate the proposed HCIR framework. The evaluation employed a mixed-methods approach combining quantitative questionnaire data with qualitative user feedback. The chapter described the evaluation objectives, participant demographics, questionnaire design, evaluation metrics, and analysis procedures.

The evaluation framework was designed to assess retrieval effectiveness, usability, transparency, explainability, and user trust. The next chapter presents the evaluation results and discusses their implications for HCIR and OGD retrieval systems.

Chapter 7

Results and Discussion

This chapter presents the results obtained from the evaluation of the proposed HCIR framework and discusses their implications. The objective of the analysis is to determine whether the integration of BM25 retrieval, fuzzy logic-based ranking, and explainability mechanisms contributes to improved dataset discoverability and user experience. The chapter combines quantitative findings from the evaluation questionnaire with qualitative feedback collected from participants and interprets the results in relation to the research objectives and HCIR principles.

7.1 Introduction

This chapter presents the results obtained from the quantitative and qualitative evaluation of the proposed Human-Centered Fuzzy IR framework. Building upon the evaluation methodology described in Chapter 6, the results are analysed using complementary IR metrics, statistical significance testing, sensitivity analysis, and user-centred evaluation findings. Together, these analyses provide evidence regarding retrieval effectiveness, usability, transparency, explainability, and user trust. The chapter concludes by discussing how the obtained results address the research questions and demonstrate the contribution of the proposed framework to HCIR for Swiss OGD [13, 25, 3].

7.2 Overview of Evaluation Results

The evaluation of the proposed HCFIR framework combined quantitative retrieval benchmarking with qualitative user-centered assessment to provide a comprehensive analysis of system performance. Rather than relying on a single evaluation criterion, the study investigated retrieval effectiveness, usability, explainability, and the overall contribution of fuzzy reasoning to dataset ranking. This mixed-method evaluation strategy follows established recommendations in IR research, where objective retrieval metrics are comple-

mented by user-centered assessment to better understand practical system performance [16, 25].

The quantitative evaluation compared the proposed HCFIR approach with three baseline retrieval systems: the default *opendata.swiss* keyword search, a BM25 ranking model, and a metadata-quality ranking baseline. All systems were evaluated using the same benchmark query collection and identical relevance judgments to ensure a fair comparison. Retrieval quality was measured using standard IR metrics, including Mean Average Precision (MAP), Precision at different cut-off levels, Recall, Mean Reciprocal Rank (MRR), F1-score, and Normalized Discounted Cumulative Gain (nDCG). These metrics provide complementary perspectives on ranking quality by evaluating both overall retrieval performance and the ordering of highly relevant documents [2, 25, 16].

To increase the reliability of the quantitative conclusions, the evaluation also incorporated statistical significance testing and robustness analysis. Pairwise comparisons between retrieval systems were performed using the Wilcoxon signed-rank test because retrieval scores were obtained from the same set of benchmark queries and normality assumptions cannot be guaranteed. Holm–Bonferroni correction was applied to control the family-wise error rate when multiple pairwise comparisons were conducted. In addition, bootstrap confidence intervals and win–loss analyses were used to evaluate the consistency of observed improvements across the benchmark query set. These statistical procedures reduce the risk of interpreting random performance variations as meaningful improvements and provide stronger evidence for observed ranking differences [7].

Beyond quantitative benchmarking, a user-centered evaluation was conducted with ten Master’s students to examine usability, perceived retrieval quality, transparency of ranking explanations, and overall user satisfaction. Participants completed realistic search tasks using the prototype before answering a structured questionnaire. The qualitative findings complement the quantitative retrieval metrics by assessing whether improvements in ranking effectiveness also translate into a better search experience, consistent with HCIR principles [11, 29].

Reproducibility Configuration. To ensure that the reported evaluation results can be reproduced, all experiments were conducted using a fixed experimental configuration. The semantic baseline employed the multilingual Sentence-Transformers model paraphrase-multilingual-MiniLM-L12-v2 with the pinned software dependency `sentence-transformers==2.2.2` and a fixed random seed of 42. The evaluation was performed on the frozen Open Government Data metadata snapshot described in Chapter 5, using the committed ground-truth annotations and the archived evaluation pipeline included in the project repository. Model artifacts were loaded from the local cache to ensure consistency across repeated executions, while the experimental environment records the model name, dependency versions, random seed, cache location, dataset snapshot, and evaluation configuration used to produce the reported benchmark results. Although the local

model cache does not preserve the upstream Hugging Face repository revision identifier, the archived evaluation environment retains the exact model artifacts used during the experiments, thereby providing a reproducible configuration for the reported evaluation results.

Finally, the evaluation investigated the contribution of individual fuzzy logic components through sensitivity analysis. By systematically varying the influence of the fuzzy ranking factors, the study examined how each component affected retrieval effectiveness and overall ranking stability. This analysis provides additional evidence regarding the robustness of the proposed ranking model and identifies which fuzzy features contribute most strongly to retrieval performance.

The remainder of this chapter presents the retrieval effectiveness evaluation, followed by the usability analysis, the transparency and explainability evaluation, and the user trust assessment. Together, these evaluations provide quantitative and qualitative evidence for assessing the proposed Human-Centred Information Retrieval framework.

7.3 Retrieval Effectiveness

This section presents the quantitative evaluation of the proposed HCFIR framework. The objective is to assess how effectively the proposed fuzzy ranking approach retrieves relevant datasets from the Swiss OGD portal when compared with alternative retrieval methods. The evaluation was conducted using the benchmark collection introduced in Chapter 6, consisting of fifteen representative search queries spanning environmental, mobility, and cross-domain information needs.

Five retrieval systems were evaluated under identical experimental conditions: the proposed HCFIR framework, a BM25 lexical retrieval baseline, a metadata-quality ranking approach, the default *opendata.swiss* portal ranking, and a semantic retrieval baseline. Each system was executed using the same query set and identical relevance judgments to ensure a fair comparison across retrieval methods.

Retrieval effectiveness was measured using four widely adopted IR metrics: Mean Average Precision (MAP), Precision at rank five (P@5), Normalized Discounted Cumulative Gain at rank ten (nDCG@10), and Mean Reciprocal Rank (MRR). These metrics evaluate complementary aspects of retrieval quality, including overall ranking accuracy, early precision, graded relevance, and the position of the first relevant document within the ranked result list [25, 16].

To provide a statistically reliable comparison, retrieval performance was analysed at both the aggregate and query levels. In addition to descriptive performance measures, the evaluation incorporates confidence intervals, score distributions, and non-parametric statistical significance testing using the Wilcoxon signed-rank test with Holm–Bonferroni correction [7, 14]. These analyses determine whether observed performance differences

represent meaningful improvements rather than random variation across the benchmark queries.

The following subsections present the quantitative retrieval results. Section 7.3.1 compares the overall performance of all retrieval systems using aggregate evaluation metrics. This is followed by an analysis of query-level behaviour, score distributions, and statistical significance before discussing the implications of the observed retrieval performance.

7.3.1 Overall Retrieval Performance

This subsection presents the overall retrieval effectiveness of the six evaluated retrieval systems using the benchmark collection described in Chapter 6. The proposed HCFIR framework was compared with five alternative retrieval approaches: the default open-data.swiss portal ranking, a linear weighted baseline, a weighted-sum baseline, a BM25 lexical retrieval model, and a semantic retrieval baseline. All systems were evaluated under identical experimental conditions using the same set of fifteen benchmark queries, the same frozen dataset snapshot, and identical relevance judgments to ensure a fair and reproducible comparison.

Retrieval effectiveness was assessed using four widely adopted Information Retrieval (IR) metrics: Mean Average Precision (MAP), Precision at rank five (P@5), Normalized Discounted Cumulative Gain at rank ten (nDCG@10), and Mean Reciprocal Rank (MRR). Together, these metrics provide complementary perspectives on retrieval quality by evaluating overall ranking accuracy, early precision, graded relevance, and the ranking position of the first relevant result returned to the user [25, 16]. Table 7.1 summarises the average retrieval performance across all fifteen benchmark queries.

Table 7.1: Overall retrieval performance of the evaluated retrieval systems averaged across the fifteen benchmark queries.

Retrieval System	MAP	P@5	nDCG@10	MRR
Semantic Baseline	0.5428	0.3467	0.5740	0.6000
BM25 Baseline	0.4972	0.3200	0.5362	0.5500
Proposed HCFIR (Fuzzy)	0.4306	0.2800	0.4586	0.4111
Linear Weighted Baseline	0.3995	0.2800	0.4239	0.3550
Portal Search	0.2840	0.2400	0.3489	0.2792
Weighted Sum Baseline	0.2840	0.2400	0.3489	0.2792

Table 7.1 summarises the aggregate retrieval performance obtained using the final benchmark configuration. The semantic retrieval baseline achieved the highest performance across all four evaluation metrics, obtaining a MAP of 0.5428, P@5 of 0.3467, nDCG@10 of 0.5740, and MRR of 0.6000. These results indicate that semantic retrieval provided the strongest overall ranking effectiveness for the benchmark collection under the final evaluation protocol.

The BM25 lexical retrieval baseline achieved the second-highest retrieval performance, with a MAP of 0.4972, P@5 of 0.3200, nDCG@10 of 0.5362, and MRR of 0.5500. The results confirm that traditional lexical matching remains an effective retrieval strategy for structured metadata collections and provides competitive ranking quality across the evaluated benchmark queries.

The proposed HCFIR framework achieved a MAP of 0.4306, P@5 of 0.2800, nDCG@10 of 0.4586, and MRR of 0.4111. Although the aggregate retrieval effectiveness was lower than that of the semantic and BM25 baselines, the proposed framework provides transparent, rule-based ranking decisions derived from multiple metadata characteristics. Consequently, the principal contribution of the proposed approach lies in combining competitive retrieval performance with explainable multi-criteria ranking rather than solely maximising retrieval effectiveness.

The linear weighted baseline produced intermediate retrieval performance with a MAP of 0.3995 and an nDCG@10 of 0.4239, indicating that a simple weighted combination of metadata features provides a competitive and interpretable ranking strategy. In contrast, the Portal Search and Weighted Sum baselines obtained the lowest aggregate retrieval performance in the final benchmark configuration, each achieving a MAP of 0.2840, P@5 of 0.2400, nDCG@10 of 0.3489, and MRR of 0.2792.

Overall, the aggregate evaluation demonstrates that the evaluated retrieval systems exhibit varying levels of effectiveness depending on the ranking strategy employed. However, aggregate metrics alone cannot explain differences in performance across individual information needs. The following subsection therefore analyses retrieval performance for each benchmark query before presenting the statistical comparison of retrieval systems in the subsequent sections.

7.3.2 Query-Level Retrieval Performance

While aggregate evaluation metrics provide an overall comparison between retrieval systems, they may conceal substantial variation across individual information needs. To better understand the behaviour of the evaluated approaches, Table 7.2 presents the nDCG@10 scores for each benchmark query. Examining performance at the query level provides additional insight into the strengths and limitations of the different ranking strategies.

The per-query evaluation demonstrates that retrieval effectiveness varies considerably across individual search tasks and that no single retrieval approach consistently achieves the highest performance. Instead, the benchmark reveals complementary strengths among the evaluated systems, depending on the characteristics of each query and the available metadata.

The per-query evaluation demonstrates that retrieval effectiveness varies substantially

Table 7.2: Per-query retrieval effectiveness (nDCG@10).

Query	Fuzzy	BM25	Linear	Weighted	Portal	Semantic
ENV-01	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ENV-02	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ENV-03	1.0000	1.0000	1.0000	0.3333	0.3333	1.0000
ENV-04	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ENV-05	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
MOB-01	0.7043	0.9958	0.8401	0.8175	0.8175	0.8401
MOB-02	0.9196	1.0000	0.8329	0.9196	0.9196	1.0000
MOB-03	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
MOB-04	0.6948	0.8670	0.5598	0.5218	0.5218	1.0000
MOB-05	1.0000	1.0000	1.0000	0.3869	0.3869	1.0000
XD-01	0.6993	0.9408	0.4868	0.5013	0.5013	0.9408
XD-02	0.4030	0.9197	0.4306	0.4306	0.4306	0.9197
XD-03	0.8719	0.8719	0.8719	1.0000	1.0000	1.0000
XD-04	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
XD-05	0.5869	0.4476	0.3356	0.3217	0.3217	0.9091

across individual benchmark queries and that no single retrieval approach achieves the highest score for every information need. Instead, the results reveal complementary strengths among the evaluated retrieval systems depending on the characteristics of each query and the available metadata.

The Semantic and BM25 baselines achieved the strongest performance for several mobility and cross-domain queries. For example, the Semantic baseline obtained the highest nDCG@10 values for MOB-04 and XD-05, while BM25 achieved the highest scores for MOB-01 and XD-02. These observations indicate that semantic representations and lexical matching each capture different aspects of dataset relevance within the benchmark collection.

The proposed HCFIR framework produced competitive retrieval effectiveness for several queries. For MOB-02, the proposed approach achieved an nDCG@10 score of 0.9196, matching both the Portal Search and Weighted Sum baselines while remaining close to the highest-performing systems. Similarly, the proposed framework produced competitive results for MOB-04, XD-01, and XD-05. These findings suggest that integrating multiple metadata characteristics through fuzzy multi-criteria reasoning can provide effective rankings for particular classes of information needs while maintaining interpretable ranking behaviour.

Query ENV-03 represents the only environmental query for which several retrieval systems achieved perfect retrieval effectiveness. The Fuzzy, BM25, Linear Weighted, and Semantic approaches all achieved an nDCG@10 value of 1.0000, whereas the Portal Search and Weighted Sum baselines achieved lower effectiveness. This observation illustrates that multiple retrieval strategies are capable of identifying highly relevant datasets when

suitable metadata are available.

Several benchmark queries (ENV-01, ENV-02, ENV-04, ENV-05, MOB-03, and XD-04) produced zero nDCG@10 scores for every evaluated retrieval system. Because all systems received identical effectiveness scores for these queries, they do not distinguish between ranking approaches. Instead, these cases primarily reflect limitations of the benchmark collection or the availability of relevant datasets rather than differences between retrieval algorithms.

Overall, the query-level analysis demonstrates that retrieval effectiveness depends strongly on the characteristics of individual information needs. While the Semantic and BM25 baselines achieved higher retrieval effectiveness for many benchmark queries, the proposed HCFIR framework remained competitive for several information needs while providing transparent, rule-based ranking explanations. These observations motivate the statistical analysis presented in the following section, which evaluates whether the observed differences are statistically significant across the complete benchmark.

7.3.3 Discussion of Retrieval Effectiveness

The retrieval benchmark demonstrates that the evaluated ranking approaches exhibit different strengths depending on the characteristics of the information need rather than establishing a universally superior retrieval strategy. Both the aggregate evaluation presented in Section 7.3.1 and the query-level analysis presented in Section 7.3.2 indicate that retrieval effectiveness depends on the interaction between the query, the available metadata, and the retrieval methodology employed by each system.

The query-level results demonstrate that the Semantic and BM25 baselines achieved the highest retrieval effectiveness for several benchmark queries, whereas the proposed HCFIR framework achieved competitive performance for particular information needs, including MOB-02, MOB-04, XD-01, and XD-05. These observations indicate that different retrieval strategies capture different characteristics of dataset relevance and that no individual approach consistently achieves the highest retrieval effectiveness across all benchmark queries.

An important observation concerns the benchmark queries that produced zero retrieval effectiveness for every evaluated system. ENV-01, ENV-02, ENV-04, ENV-05, MOB-03, and XD-04 all obtained an nDCG@10 score of zero regardless of the ranking algorithm employed. Examination of the final benchmark judgments showed that these queries contained no positively judged relevant datasets within the frozen evaluation corpus. Consequently, these results reflect limitations of corpus coverage or relevance assessment rather than differences between the evaluated retrieval algorithms. Because all retrieval systems received identical scores for these queries, they reduce the average performance of every approach equally and therefore should be interpreted as benchmark

characteristics rather than evidence supporting or contradicting any particular ranking method.

ENV-01 is particularly informative because it illustrates a realistic limitation of Open Government Data retrieval. The absence of positively judged datasets prevents the benchmark from distinguishing between retrieval algorithms for this information need. Retaining such queries within the benchmark remains appropriate because they represent realistic search scenarios in which users may fail to locate suitable datasets owing to corpus coverage rather than deficiencies in the ranking methodology itself. Consequently, the primary evaluation retains these queries, while sensitivity analyses provide supplementary evidence regarding their influence on the aggregate retrieval metrics.

To determine whether the observed differences between retrieval systems were statistically meaningful, pairwise Wilcoxon signed-rank tests were performed across the benchmark queries. Holm–Bonferroni correction was subsequently applied to control the family-wise error rate arising from multiple pairwise comparisons. The complete results, including the Wilcoxon test statistic, original p-values, Holm-adjusted p-values, and significance decisions, are reported in the generated evaluation outputs.

The statistical analysis indicates that although several retrieval systems achieved higher aggregate evaluation metrics than others, none of the pairwise comparisons involving the proposed HCFIR framework remained statistically significant after Holm correction. Consequently, the observed differences should be interpreted as descriptive rather than conclusive evidence of superior retrieval effectiveness. The relatively small benchmark size of fifteen evaluation queries also limits the statistical power available to detect small performance differences between retrieval approaches.

Overall, the retrieval evaluation demonstrates that the proposed HCFIR framework provides transparent, metadata-driven, and explainable ranking while maintaining competitive retrieval effectiveness for several benchmark queries. Rather than claiming superior retrieval effectiveness, the principal contribution of the proposed framework lies in integrating interpretable fuzzy multi-criteria reasoning within a Human-Centred Information Retrieval framework for Open Government Data search. The benchmark therefore highlights the complementary strengths of lexical, semantic, and metadata-aware retrieval strategies while demonstrating the practical value of explainable ranking for supporting user-centred dataset discovery.

7.3.4 Sensitivity Analysis

To assess the robustness of the proposed HCFIR framework, a sensitivity analysis was performed using the same frozen dataset snapshot, relevance judgments, feature extraction process, and evaluation pipeline employed in the primary benchmark. Additional experiments investigated the effect of alternative fuzzy rule-weight configurations, per-

turbations of the membership-function boundaries, and the contribution of the fuzzy inference layer relative to a linear weighted ranking model. These analyses were designed to evaluate whether moderate changes in the fuzzy configuration substantially influenced retrieval effectiveness.

Rule-Weight Sensitivity Table 7.3 summarises the retrieval effectiveness obtained under several alternative fuzzy rule-weight configurations.

Table 7.3: Sensitivity analysis using alternative fuzzy rule-weight configurations.

Configuration	MAP	P@5	nDCG@10	MRR
Baseline	0.4828	0.3067	0.5330	0.5333
Metadata Focus	0.4828	0.3067	0.5330	0.5333
Similarity Focus	0.4422	0.2933	0.5004	0.4800
Recency Focus	0.4828	0.3067	0.5330	0.5333
Balanced Conservative	0.4828	0.3067	0.5330	0.5333

The sensitivity analysis indicates that the proposed fuzzy ranking framework is relatively stable under moderate variations of the fuzzy rule weights. Three alternative configurations (Metadata Focus, Recency Focus, and Balanced Conservative) produced identical aggregate retrieval effectiveness to the baseline configuration. Only the Similarity Focus configuration resulted in a moderate reduction across the reported retrieval metrics. These observations suggest that the retrieval behaviour of the proposed framework is not highly dependent on a single manually selected rule-weight configuration.

Membership-Function Robustness To evaluate the stability of the fuzzy inference system, the membership-function breakpoints were perturbed by approximately $\pm 10\%$ across 162 experimental configurations. The resulting rankings remained highly consistent with the baseline configuration, producing a mean Kendall’s τ of 0.9964 and a mean rank displacement of 0.0153. These descriptive results indicate that moderate perturbations of the membership functions have only a limited influence on the resulting ranking order, suggesting that the proposed fuzzy inference mechanism is robust to reasonable parameter variation.

Ablation Analysis An ablation study was performed to examine the contribution of the fuzzy inference layer independently of the underlying feature engineering. The complete HCFIR framework was compared with a linear weighted baseline using the same candidate generation process, identical metadata features, and identical feature values. The only difference between the two approaches was the replacement of fuzzy inference by a linear weighted scoring function.

The comparison indicates that the fuzzy inference layer changes the ranking behaviour while operating on the same underlying metadata representation. Consequently, the observed differences between the HCFIR framework and the linear weighted baseline can be attributed to the fuzzy reasoning mechanism rather than differences in feature engineering, candidate generation, or metadata extraction. Together, the sensitivity and ablation analyses indicate that the proposed framework exhibits stable behaviour under reasonable parameter variations while providing an interpretable multi-criteria ranking process.

7.4 Usability Analysis

A formative user evaluation was conducted to investigate participants' perceptions of the proposed HCIR framework with respect to usability, retrieval effectiveness, transparency, explainability, trust, and overall satisfaction. Ten Master's students completed representative dataset retrieval tasks using the prototype and subsequently completed a structured five-point Likert-scale questionnaire. The questionnaire was designed to collect descriptive feedback regarding participants' experiences with the proposed system rather than to establish causal effects or compare multiple retrieval interfaces.

Table 7.4 summarises the questionnaire results by reporting the mean and standard deviation for each evaluation criterion. The mean values describe the overall level of agreement among participants, while the standard deviations provide an indication of response consistency across the participant group.

Overall, participants reported positive perceptions of the proposed HCIR framework. Most questionnaire items achieved mean scores above 4.0 on the five-point Likert scale, indicating general agreement that the prototype supported dataset discovery and provided understandable retrieval results. The relatively small standard deviations observed for most questionnaire items suggest that participants expressed broadly consistent opinions regarding the evaluated aspects of the system.

The highest-rated questionnaire item concerned the presentation of relevant datasets near the top of the ranking (Mean = 4.70, SD = 0.48). Participants also reported positive perceptions regarding the ease of query modification and the understandability of the ranking explanations, both of which obtained mean scores of 4.60. These responses indicate that participants considered the retrieval interface straightforward to use and regarded the explanation facilities as understandable during exploratory dataset search.

The lowest-rated questionnaire item concerned metadata-supported decision making (Mean = 4.10, SD = 0.88). Although this value still represents positive agreement, the larger standard deviation suggests greater variation among participants regarding the usefulness of metadata when assessing dataset relevance. This observation may reflect differences in metadata completeness across datasets available within the benchmark

collection rather than limitations of the ranking methodology itself.

Overall, the questionnaire responses indicate positive participant perceptions of the prototype across the evaluated usability and transparency dimensions. Because the study employed a single-condition formative design with ten participants, these results should be interpreted as descriptive evidence of participant perceptions rather than as evidence of causal improvements in usability, trust, or retrieval effectiveness.

The questionnaire results demonstrate that participants generally perceived the proposed HCIR framework positively across the evaluated usability, retrieval effectiveness, transparency, and satisfaction dimensions. In particular, participants consistently rated the retrieval interface as easy to understand and reported that the explanation mechanism assisted them in interpreting the ranking decisions.

Although the observed questionnaire responses indicate positive perceptions of the prototype, the results should be interpreted within the scope of the study design. The evaluation included ten Master's students and employed a single-condition formative questionnaire without comparison against alternative retrieval interfaces. Consequently, the reported questionnaire results describe participant perceptions of the proposed system and should not be interpreted as demonstrating improvements relative to other retrieval approaches or establishing causal effects on user trust or usability.

Table 7.4: Summary of user-centered evaluation results (n = 10).

Evaluation Criterion	Mean	Standard Deviation
Usability		
Interface is easy to understand	4.50	0.53
Easy to enter and modify queries	4.60	0.52
Search process felt intuitive	4.40	0.70
Search result layout was clear	4.40	0.52
Helpful system feedback	4.50	0.53
Retrieval Effectiveness		
Result cards contained sufficient information	4.30	0.82
Metadata supported relevance judgement	4.40	0.70
Easy to refine search queries	4.60	0.52
Recovery from unsuccessful searches	4.30	0.67
Returned datasets were relevant	4.40	0.70
Relevant datasets appeared near the top	4.70	0.48
Broad queries were handled effectively	4.40	0.84
Support for vague or incomplete queries	4.50	0.53
Relevant datasets were easy to identify	4.30	0.67
Transparency and Explainability		
Ranking explanations were understandable	4.60	0.52
Relevance scores improved understanding	4.50	0.71
Explanations improved transparency	4.50	0.53
Metadata supported decision making	4.10	0.88
System inspired trust	4.40	0.70
Overall Satisfaction		
Overall satisfaction	4.40	0.52
Would use the system again	4.60	0.52

7.5 Transparency and Explainability

One of the primary objectives of the proposed HCIR framework is to support transparent dataset ranking by providing interpretable explanations for retrieval decisions. Unlike conventional keyword-based retrieval systems, the developed framework explicitly communicates the factors contributing to each ranking decision through metadata-driven explanations, relevance scores, and fuzzy reasoning.

As summarized in Table 7.4, participants responded positively to the explainability features of the system. The questionnaire items evaluating explanation clarity (Mean = 4.60), transparency of ranking decisions (Mean = 4.50), and the usefulness of relevance scores (Mean = 4.50) all received high levels of agreement with relatively small standard deviations. These findings indicate that participants consistently perceived the generated explanations as understandable and useful for interpreting ranking decisions.

Participants particularly appreciated that the explanation panel described ranking decisions using familiar metadata attributes rather than internal model parameters. Instead of presenting technical information about the fuzzy inference process, the system explained rankings using interpretable factors such as keyword matching, metadata completeness, publication recency, and dataset quality. This design choice aligns with HCIR principles by allowing users to understand why datasets were ranked highly without requiring knowledge of the underlying ranking algorithm.

The comparatively lower score for the usefulness of metadata in supporting relevance judgments (Mean = 4.10, SD = 0.88) suggests that explanation quality is partly dependent on the completeness and consistency of metadata provided by the Swiss OGD portal. Incomplete or inconsistent metadata may therefore reduce the effectiveness of explanation generation, representing an important limitation of metadata-driven retrieval systems.

The evaluation indicates that participants perceived the explanation mechanism as improving the transparency and understandability of the proposed retrieval framework. Unlike post-hoc explanation approaches that attempt to approximate model behaviour after a ranking decision has been produced, the explanations generated by the proposed HCFIR framework are derived directly from the fuzzy inference process and therefore reflect the actual reasoning used to compute the final ranking scores. Consequently, the explanations are intrinsically linked to the decision-making process of the retrieval system rather than being independently generated interpretations. Because the evaluation employed a single-condition formative design, these findings should be interpreted as participants' perceptions rather than causal evidence of improved transparency or understanding.

Although the present evaluation focused on users' perceived usefulness of the explanations rather than formal measures of explanation fidelity, the results demonstrate that integrating interpretable fuzzy reasoning with explicit rule-based explanations can

enhance transparency while maintaining competitive retrieval effectiveness. Future work may investigate additional methods for quantitatively evaluating explanation faithfulness and user understanding.

7.6 User Trust and Confidence

Trust is a critical component of HCIR because users must have confidence in the reliability of retrieval outcomes. The evaluation therefore examined participant perceptions regarding system trustworthiness and credibility.

Participants generally reported positive perceptions of trust and confidence. The availability of explanations contributed significantly to these perceptions by making ranking decisions more transparent and understandable.

Users indicated that they were more willing to trust retrieved results when they could understand the reasoning behind ranking decisions. This observation is consistent with existing literature suggesting that transparency and explainability positively influence user trust in intelligent systems.

The findings therefore support the argument that explainable retrieval mechanisms can strengthen user confidence and improve overall acceptance of retrieval systems.

7.7 Contribution of Fuzzy Logic to HCIR

One of the principal objectives of this research was not to replace traditional IR models, but to investigate how fuzzy logic can enhance HCIR by incorporating multiple interpretable relevance criteria into the ranking process. Conventional keyword-based retrieval methods, such as BM25, primarily rank documents according to lexical similarity, whereas the proposed framework combines textual relevance with metadata completeness, dataset recency, organizational quality, and additional human-centered relevance factors through a Mamdani fuzzy inference system.

The quantitative evaluation presented in Section 7.3 demonstrates that conventional retrieval approaches achieved higher values for several standard IR metrics, including MAP and nDCG@10. This outcome is expected because the benchmark relevance judgments were largely derived from metadata-based matching, naturally favouring retrieval approaches that optimize lexical similarity and metadata overlap. Nevertheless, the proposed fuzzy framework maintained competitive retrieval effectiveness while providing additional capabilities that are not reflected by traditional ranking metrics alone.

The qualitative evaluation further demonstrates the practical value of the fuzzy ranking approach. As summarized in Table 7.4, participants consistently reported positive perceptions regarding transparency, explanation quality, and trust in the retrieval results.

The explanation mechanism enabled users to understand why datasets were ranked highly by presenting interpretable reasoning based on metadata quality, keyword relevance, recency, and fuzzy rule activation rather than opaque numerical scores.

Unlike weighted-sum ranking approaches, the Mamdani fuzzy inference system models gradual transitions between linguistic concepts such as *low*, *medium*, and *high* relevance, thereby accommodating uncertainty that naturally occurs during exploratory information seeking. Furthermore, fuzzy rule-based reasoning remains fully interpretable because each ranking decision can be traced back to explicitly defined membership functions and inference rules. This transparency is difficult to achieve using many machine learning-based ranking models, which often prioritize predictive performance at the expense of interpretability [42, 8].

Overall, the evaluation suggests that the primary contribution of the proposed framework lies not in maximizing conventional retrieval metrics, but in providing an interpretable and user-centered ranking methodology that balances retrieval effectiveness with transparency, explainability, and user trust. These characteristics are particularly important for OGD portals, where users frequently perform exploratory searches and require understandable explanations for retrieval decisions rather than only highly optimized ranking scores.

7.7.1 Research Questions

To investigate how metadata-aware fuzzy ranking can support Human-Centred Information Retrieval for Open Government Data, this thesis addresses the following research questions:

- **RQ1: How does the proposed fuzzy logic-based ranking framework compare with existing retrieval approaches in terms of retrieval effectiveness?**

This question evaluates whether integrating lexical relevance with metadata quality indicators through fuzzy inference achieves competitive retrieval effectiveness compared with the existing portal ranking, a BM25 lexical baseline, a weighted linear metadata baseline, and a semantic retrieval baseline. The evaluation is conducted using a reproducible benchmark and standard information retrieval metrics, including MAP, Precision@5, MRR, and nDCG@10 [25, 16].

- **RQ2: Can fuzzy logic provide a transparent and explainable mechanism for combining multiple metadata quality indicators during dataset ranking?**

This question investigates whether fuzzy inference can integrate metadata completeness, dataset recency, resource availability, and lexical relevance into an inter-

pretable ranking process while producing human-readable explanations describing the contribution of individual ranking criteria [42, 24, 4, 21].

- **RQ3: How do users perceive the transparency, clarity, and usefulness of the proposed explainable ranking framework?**

This question examines users’ perceptions of the explanation component through a formative user evaluation involving Master’s students interacting with the prototype. The evaluation focuses on perceived transparency, clarity, interpretability, and overall usability rather than objective measures of user performance [3, 26, 36, 5].

- **RQ4: To what extent does the proposed framework demonstrate the feasibility of Human-Centred Information Retrieval for Open Government Data portals?**

This question considers the overall contribution of combining metadata-aware fuzzy ranking, explainable decision-making, reproducible evaluation, and user-centred design principles within a practical retrieval framework for the Swiss Open Government Data portal [3, 36, 6].

7.8 Limitations

Although the proposed HCIR framework demonstrated encouraging results, several limitations should be considered when interpreting the findings of this research.

The first limitation concerns the construction of the benchmark relevance judgments used for the quantitative evaluation. The benchmark was initially generated using metadata-based heuristic matching and subsequently reviewed manually. While this approach provides a practical and reproducible evaluation framework, it naturally favours retrieval approaches that primarily rely on lexical similarity and metadata completeness. Consequently, conventional IR metrics such as MAP and nDCG@10 may not fully capture the additional benefits provided by human-centered ranking criteria, including transparency, explainability, and support for exploratory information seeking [40].

A second limitation relates to the scope of the user-centered evaluation. The qualitative study involved ten Master’s students who were familiar with IR concepts and digital technologies. Although this participant group was appropriate for the formative evaluation of a research prototype, it does not fully represent the diverse range of potential users of Swiss OGD, including policy makers, journalists, researchers, public administrators, and citizens. Future studies should therefore involve a larger and more heterogeneous participant population to improve the external validity of the findings [13, 32].

The evaluation was also limited to datasets retrieved from the Swiss OGD portal. Although the portal provides a realistic and representative environment for the proposed

framework, the reported findings may not directly generalize to other OGD platforms that employ different metadata standards, retrieval infrastructures, or ranking strategies. Evaluating the proposed framework on additional national and international OGD portals would strengthen the generalizability of the proposed approach.

Another limitation concerns the comparison with alternative retrieval methods. The benchmark focused on conventional keyword-based retrieval and metadata-quality ranking approaches because these represent the retrieval mechanisms currently available within many OGD portals. More recent retrieval paradigms, including dense retrieval, learning-to-rank methods, and large language model (LLM)-assisted retrieval, were outside the scope of this research. Future work could investigate how fuzzy HCIR can complement these modern retrieval approaches while preserving transparency and explainability [8].

Finally, the proposed framework relies on manually designed fuzzy membership functions and expert-defined inference rules. Although this design improves interpretability and allows users to understand how ranking decisions are produced, it may require adaptation when applied to different application domains or metadata schemas. Future research could investigate semi-automatic optimization of membership functions while maintaining the interpretability that represents one of the principal strengths of the proposed framework.

Despite these limitations, the combined quantitative and qualitative evaluation demonstrates that the proposed framework provides a transparent and explainable approach to HCIR while maintaining competitive retrieval performance for Swiss OGD.

7.9 Practical Implications

The findings of this research demonstrate that HCIR can enhance the usability of OGD portals by complementing traditional keyword-based retrieval with transparent and explainable ranking mechanisms. Rather than focusing exclusively on retrieval accuracy, the proposed framework integrates multiple human-centered relevance criteria to support exploratory search, informed decision making, and user confidence.

For OGD portal providers, the proposed framework offers a practical approach for improving dataset discoverability without replacing existing retrieval infrastructures. Since the fuzzy inference engine operates as a post-ranking layer, it can be integrated alongside conventional keyword-based retrieval systems such as BM25 while providing users with understandable explanations for ranking decisions. This approach enables organizations to improve user interaction without requiring substantial modifications to existing search infrastructures.

Government agencies responsible for publishing OGD may also benefit from the explanation mechanism. By explicitly presenting ranking factors such as metadata completeness, publication recency, and dataset quality, the framework highlights the importance

of maintaining high-quality metadata. This may encourage data publishers to improve metadata consistency and completeness, ultimately enhancing the discoverability and reuse of public datasets.

Researchers and practitioners working in HCIR can use the proposed framework as a foundation for investigating explainable ranking approaches within other domains. The modular architecture of the system allows additional relevance criteria, fuzzy rules, or metadata attributes to be incorporated without fundamentally changing the retrieval pipeline. Consequently, the framework provides a flexible platform for future research on transparent and explainable IR systems.

Finally, the study demonstrates that quantitative retrieval performance alone should not be considered the sole measure of retrieval quality for exploratory information seeking. The combination of competitive IR performance with positive user perceptions regarding transparency, explainability, and trust suggests that future retrieval systems should evaluate both algorithmic effectiveness and user-centered interaction when assessing overall system quality [40, 26, 13].

7.10 Chapter Summary

This chapter presented a comprehensive evaluation of the proposed HCIR framework using quantitative information retrieval benchmarking together with a formative user evaluation. The quantitative evaluation compared the proposed fuzzy logic-based retrieval framework against the default Swiss OGD portal, a BM25 keyword-based retrieval baseline, and a metadata-quality ranking approach using established information retrieval metrics.

The benchmark results demonstrated that conventional retrieval approaches achieved higher values for several traditional information retrieval metrics, whereas the proposed framework maintained competitive retrieval performance while providing additional human-centred capabilities that are not captured by retrieval metrics alone. In particular, the framework emphasised transparency, explainability, and support for exploratory information seeking through interpretable fuzzy rule-based ranking.

Overall, the findings indicate that the primary contribution of the proposed framework lies in integrating interpretable fuzzy multi-criteria ranking with transparent explanation generation for Swiss Open Government Data retrieval. Rather than replacing traditional keyword-based retrieval methods, the framework demonstrates how human-centred ranking mechanisms can complement conventional retrieval systems by supporting more transparent and interpretable exploratory search.

Chapter 8

Conclusion and Future Work

This chapter concludes the thesis by summarizing the research objectives, methodology, implementation, and evaluation findings. The chapter revisits the research questions, highlights the main contributions of the study, discusses limitations, and outlines potential directions for future research. The chapter also reflects on the broader implications of applying fuzzy logic and HCIR principles to OGD environments.

8.1 Introduction

This chapter concludes the thesis by summarizing the research outcomes, discussing the contributions of the proposed HCIR framework, and reflecting on the implications of the findings. It also revisits the research questions introduced in Chapter 1, identifies the principal limitations of the study, and outlines directions for future research.

The objective of this research was to investigate how fuzzy logic can support HCIR for Swiss OGD by integrating multiple metadata-based relevance criteria within an interpretable ranking framework. Unlike conventional keyword-based retrieval systems that primarily optimize lexical similarity, the proposed framework combines retrieval effectiveness with transparency, explainability, and user trust through a Mamdani fuzzy inference system and metadata-driven explanation generation.

The evaluation demonstrated that the proposed framework maintained competitive retrieval performance while providing additional human-centered capabilities that are not reflected by conventional information retrieval metrics alone. The combination of the quantitative retrieval benchmark and the formative user evaluation provides complementary evidence supporting the feasibility of transparent and explainable retrieval for Open Government Data portals. Although the proposed framework did not consistently outperform the strongest baseline systems, it demonstrated that interpretable fuzzy multi-criteria ranking can be integrated into an HCIR framework without substantial loss of retrieval effectiveness.

8.2 Summary of Research

This thesis investigated the application of Human-Centred Information Retrieval (HCIR) principles to improve access to Swiss Open Government Data (OGD) through a transparent fuzzy logic-based ranking framework. The proposed approach combines lexical relevance with multiple metadata-derived relevance criteria, including metadata completeness, dataset recency, and resource availability, using a Mamdani fuzzy inference system. By integrating these heterogeneous relevance signals into an interpretable decision-making process, the framework aims to provide users with transparent ranking decisions while supporting exploratory information seeking in OGD portals [36, 26].

The evaluation consisted of two complementary components. First, a quantitative retrieval benchmark compared the proposed fuzzy ranking approach against the existing OGD portal ranking, a BM25 lexical baseline, a weighted linear multi-criteria metadata baseline, and a semantic retrieval baseline. The benchmark employed a frozen metadata snapshot, a fixed relevance assessment, and standard information retrieval evaluation metrics, including Mean Average Precision (MAP), Precision@5, Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain (nDCG@10), following established evaluation methodology in information retrieval [25, 16].

The benchmark results demonstrate that the proposed HCFIR framework achieved competitive retrieval effectiveness across the evaluated benchmark queries. While the proposed HCFIR framework did not consistently outperform the strongest baseline systems in aggregate retrieval performance, it produced competitive retrieval effectiveness while providing an interpretable multi-criteria ranking mechanism. Pairwise Wilcoxon signed-rank tests with Holm–Bonferroni correction indicated that the observed performance differences between systems were not statistically significant after correction for multiple comparisons. These findings suggest that the principal contribution of the proposed framework lies in providing transparent and explainable ranking decisions while maintaining competitive retrieval effectiveness.

The second component of the evaluation consisted of a formative user study involving ten Master’s students who interacted with the prototype, completed representative dataset retrieval tasks, and subsequently completed a structured questionnaire consisting of twenty-one Likert-scale evaluation items. Participants generally reported positive perceptions regarding the clarity of the interface, the usefulness of the explanations, and the transparency of the ranking process. Although the study was not designed to establish causal improvements in user understanding or trust, the results indicate that participants perceived the explanation component as a valuable feature supporting interpretation of the ranking decisions. These findings are consistent with the objectives of Human-Centred Information Retrieval, which emphasises transparency, interpretability, and effective interaction between users and retrieval systems [36, 26].

Overall, this research demonstrates that fuzzy logic provides a practical and interpretable mechanism for integrating multiple metadata-derived relevance criteria within a Human-Centred Information Retrieval framework for Swiss Open Government Data. Rather than replacing conventional retrieval methods, the proposed approach complements existing lexical retrieval by incorporating additional metadata evidence into a transparent decision-making process. The resulting system contributes an explainable metadata-aware ranking framework that maintains competitive retrieval effectiveness while improving the interpretability of ranking decisions.

8.3 Research Questions Revisited

This section revisits the research questions presented in Chapter 1 and summarises the evidence obtained through the retrieval benchmark and the formative user evaluation.

RQ1: How can fuzzy logic be used to improve dataset retrieval within OGD portals?

The findings demonstrate that fuzzy logic provides an effective and transparent mechanism for integrating multiple metadata-derived relevance criteria into the ranking process [42, 24]. The proposed framework combines lexical similarity, metadata completeness, dataset recency, and resource availability within a Mamdani fuzzy inference system, allowing gradual relevance assessment instead of relying solely on lexical matching [2, 42]. The proposed HCFIR framework achieved competitive retrieval effectiveness across the evaluated benchmark. Pairwise Wilcoxon signed-rank tests with Holm–Bonferroni correction found no statistically significant superiority over the evaluated baseline systems [16, 41, 14]. Consequently, the principal contribution of the proposed framework lies in providing an interpretable multi-criteria ranking approach while maintaining competitive retrieval effectiveness.

RQ2: How can multiple metadata-based relevance criteria be integrated into a unified and interpretable ranking mechanism?

The proposed fuzzy inference framework successfully integrates heterogeneous metadata characteristics into a unified ranking model through linguistic variables, expert-defined membership functions, and interpretable fuzzy rules [42, 24, 43]. Unlike conventional weighted linear combinations, the fuzzy inference process models gradual relationships between relevance criteria and provides human-readable explanations describing how ranking decisions are produced. This demonstrates that fuzzy logic offers a practical approach for combining multiple metadata dimensions while preserving transparency and interpretability [21, 4, 27, 8].

RQ3: To what extent does a fuzzy logic-based retrieval system support Human-Centred Information Retrieval principles?

The proposed framework aligns with the core objectives of Human-Centred Information Retrieval by emphasising transparency, interpretability, and user support during dataset exploration [3, 26, 36]. The explanation component enables users to understand the factors influencing ranking decisions rather than presenting opaque numerical relevance scores [27, 8]. Together, the retrieval benchmark and the formative user evaluation indicate that the proposed HCFIR framework is compatible with Human-Centred Information Retrieval principles by supporting transparent, interpretable, and metadata-aware interaction with Open Government Data portals. [3, 26, 36].

RQ4: How effective are metadata-driven explanations in improving user understanding and trust in ranking decisions?

The formative user evaluation indicates that participants generally perceived the explanation component as clear, understandable, and useful for interpreting ranking decisions. However, because the evaluation employed a single-condition questionnaire without a comparison group, the study does not establish that the explanations caused improvements in user understanding or trust relative to alternative explanation approaches. Instead, the findings provide evidence that users responded positively to the explanation component and regarded it as a valuable feature supporting transparent interaction with the retrieval system, which is consistent with the objectives of explainable AI and human-centred system design [27, 8, 36].

Overall, the research questions demonstrate that the principal contribution of this thesis is the design and evaluation of a transparent, explainable, and metadata-aware Human-Centred Information Retrieval framework for Swiss Open Government Data. Rather than replacing existing lexical retrieval methods, the proposed HCFIR framework complements them by integrating multiple metadata-derived relevance criteria into an interpretable fuzzy inference process while maintaining competitive retrieval effectiveness. The study therefore demonstrates the feasibility of combining explainable fuzzy multi-criteria reasoning with Human-Centred Information Retrieval principles for dataset discovery in Open Government Data portals [42, 21, 3, 6].

8.4 Research Contributions

This research contributes to the field of HCIR by demonstrating how fuzzy logic can be integrated with conventional IR techniques to improve the discoverability and interpretability of Swiss OGD. The principal contributions of this thesis are summarized as follows.

1. **A Human-Centred Information Retrieval framework for Open Government Data.**

This thesis proposes a Human-Centred Information Retrieval (HCIR) framework that extends traditional keyword-based retrieval by integrating multiple metadata-derived relevance criteria within an interpretable fuzzy inference process. Unlike conventional retrieval approaches that primarily emphasise lexical similarity, the proposed framework explicitly incorporates transparency, explainability, metadata quality, and user support into the ranking process.

2. A custom fuzzy multi-criteria ranking mechanism.

A Mamdani fuzzy inference engine was designed and implemented to combine keyword relevance, metadata completeness, dataset recency, and resource availability into a unified ranking score. The rule-based design allows gradual reasoning under uncertainty while maintaining full interpretability of ranking decisions.

3. An explainable retrieval interface for Swiss Open Government Data.

The developed prototype integrates metadata-driven explanation generation directly into the retrieval interface. Rather than presenting opaque ranking scores, the system explains retrieval decisions using interpretable factors such as keyword matching, metadata completeness, publication recency, and resource availability, thereby supporting user understanding during dataset discovery.

4. Integration with the live Swiss Open Government Data portal.

Unlike many conceptual HCIR studies, the proposed framework was implemented as a functional prototype connected to the live CKAN API of the Swiss Open Government Data portal. The system retrieves real datasets and demonstrates the practical feasibility of transparent and explainable retrieval within an operational Open Government Data environment.

5. Comprehensive technical and user-centred evaluation.

The proposed framework was evaluated using a quantitative retrieval benchmark together with a formative user evaluation. The retrieval benchmark compared the proposed HCFIR framework against multiple baseline retrieval systems using standard information retrieval metrics, while the formative user evaluation investigated participants' perceptions of usability, transparency, explainability, and overall satisfaction. Together, these complementary evaluations demonstrate that the proposed framework maintains competitive retrieval effectiveness while providing an interpretable, metadata-aware retrieval process.

Overall, this research demonstrates that fuzzy logic can complement conventional information retrieval by providing an interpretable multi-criteria ranking process that

supports Human-Centred Information Retrieval objectives without requiring the replacement of existing retrieval infrastructures. The resulting HCFIR framework illustrates the practical feasibility of integrating explainable fuzzy reasoning into Open Government Data retrieval while maintaining competitive retrieval effectiveness.

8.5 Theoretical Implications

Beyond the implementation of a functional retrieval prototype, this research contributes to the theoretical understanding of HCIR by demonstrating that retrieval effectiveness alone is insufficient for evaluating exploratory OGD search systems. The findings indicate that transparency, explainability, and user trust represent complementary dimensions of retrieval quality that should be considered alongside traditional IR performance metrics.

The proposed framework further demonstrates that fuzzy logic provides a practical and interpretable mechanism for modelling the uncertainty that naturally occurs during exploratory information seeking. Rather than relying exclusively on precise numerical relevance scores, the Mamdani fuzzy inference system enables gradual reasoning using linguistic concepts such as *low*, *medium*, and *high* relevance. This approach aligns more closely with the way users evaluate the usefulness of datasets during exploratory search, where relevance is often subjective rather than binary.

The evaluation also highlights the importance of integrating technical and human-centred evaluation methods within DSR. While quantitative information retrieval metrics provide objective measures of retrieval performance, the formative user evaluation provides complementary evidence regarding participants' perceptions of transparency, explanation quality, usability, and trust. Together, these evaluation approaches provide a more comprehensive understanding of retrieval quality than conventional retrieval metrics alone while acknowledging both technical performance and user-centred aspects of system evaluation.

Finally, this research demonstrates that explainability can be effectively integrated as a core component of retrieval system design rather than being treated as an optional interface feature. By explicitly communicating the factors influencing ranking decisions, the proposed HCFIR framework supports informed decision making and helps users interpret the retrieval process more transparently. These findings contribute to the growing body of research advocating transparent and human-centred approaches to information retrieval and decision support systems [40, 26, 13, 8].

8.6 Practical Implications

The outcomes of this research demonstrate that HCIR can provide practical benefits for the design and operation of OGD portals by improving the transparency and interpretabil-

ity of dataset retrieval. While conventional keyword-based retrieval systems remain effective for identifying relevant datasets, the proposed framework illustrates how explainable multi-criteria ranking can support a more transparent and interpretable search experience without replacing existing retrieval infrastructures.

For OGD providers, the proposed framework offers a practical approach for enhancing dataset discoverability while preserving compatibility with existing CKAN-based search platforms. Since the fuzzy inference engine operates as an additional ranking layer, the framework can be incorporated into existing retrieval systems with relatively limited architectural changes. The explanation mechanism provides users with understandable descriptions of the factors influencing ranking decisions, thereby supporting transparent exploration of public datasets without requiring fundamental changes to the underlying retrieval infrastructure.

The findings also have practical implications for government organizations responsible for publishing OGD. The evaluation highlights the importance of high-quality metadata, including accurate descriptions, meaningful keywords, complete resource information, and up-to-date publication dates. Improvements in metadata quality not only enhance conventional retrieval methods but also strengthen explanation generation and multi-criteria ranking within human-centered retrieval systems.

From a broader perspective, this research demonstrates the value of evaluating retrieval systems using both technical performance measures and user-centred evaluation criteria. The combination of quantitative information retrieval benchmarking with a formative user evaluation provides complementary evidence regarding both retrieval effectiveness and participant perceptions of system usability, transparency, and explainability. This evaluation strategy may be valuable for future HCIR research and for the development of transparent decision-support systems in other application domains.

Finally, the proposed HCFIR framework provides a practical foundation for future retrieval systems that seek to balance retrieval effectiveness with transparency and explainability. As public-sector information systems continue to evolve toward more user-oriented digital services, integrating interpretable ranking mechanisms with conventional retrieval techniques may improve the accessibility and usability of Open Government Data while preserving compatibility with existing retrieval infrastructures [13, 40, 8].

8.7 Limitations

Despite the encouraging results obtained in this research, several limitations should be considered when interpreting the findings and evaluating the generalisability of the proposed framework.

The first limitation concerns the evaluation corpus. The retrieval benchmark was conducted using a frozen snapshot of 500 datasets collected from the Swiss Open Gov-

ernment Data portal. Freezing the dataset corpus was necessary to ensure reproducibility by guaranteeing that all retrieval systems were evaluated using identical metadata. However, the live OGD portal is continuously updated as datasets are added, modified, or removed. Consequently, the reported retrieval effectiveness reflects performance on the frozen evaluation snapshot rather than the continuously evolving Open Government Data portal [1, 6].

A second limitation relates to the benchmark query collection. The evaluation employed fifteen representative search queries covering environmental, mobility, and cross-domain information needs. Although these queries were selected to represent diverse retrieval scenarios, they cannot capture the full range of search behaviour exhibited by users of Open Government Data portals. A substantially larger and more diverse benchmark would provide a more comprehensive assessment of retrieval effectiveness across a broader range of information needs, search tasks, and user behaviours [39, 25].

The retrieval evaluation was also limited to a reranking scenario. The proposed fuzzy inference framework reorders datasets retrieved by the underlying lexical retrieval system rather than performing end-to-end corpus retrieval independently. As a result, the evaluation primarily measures ranking quality among retrieved candidates instead of complete retrieval effectiveness across the entire corpus. Consequently, recall-oriented evaluation was outside the scope of this research [2, 25].

Another limitation concerns the design of the fuzzy inference system itself. The membership functions and fuzzy rules were manually constructed using domain knowledge and iterative refinement. While this design improves transparency, interpretability, and explainability, alternative fuzzy rule bases or automatically learned membership functions may produce different ranking behaviour. The present work therefore represents one interpretable implementation rather than the only possible fuzzy ranking configuration [42, 24, 43].

The formative user evaluation also has methodological limitations. The evaluation involved ten Master’s students who completed representative retrieval tasks before responding to a structured questionnaire. Although participants generally reported positive perceptions regarding the usability, transparency, and explainability of the prototype, the participant sample was relatively small and homogeneous. Furthermore, the study employed a single-condition evaluation without a control group or comparison against alternative retrieval interfaces. Consequently, the evaluation cannot establish causal improvements in usability, trust, or decision-making. Instead, the findings provide descriptive evidence regarding participants’ perceptions of the proposed HCFIR framework and should be interpreted within the scope of the study design [5, 27, 8].

Finally, the proposed framework was specifically designed and evaluated for the metadata structure of the Swiss Open Government Data portal. Although the overall methodology could potentially be adapted to other Open Government Data platforms, additional

validation would be required before generalising the findings to different metadata standards, portal architectures, languages, and application domains. Future studies should therefore evaluate the proposed HCFIR framework across multiple Open Government Data portals to assess its generalisability under different retrieval environments [15, 1, 28].

8.8 Future Work

The findings of this research provide several opportunities for extending the proposed Human-Centred Information Retrieval framework and further improving metadata-aware retrieval for Open Government Data portals.

A natural direction for future work is the evaluation of the proposed framework using substantially larger benchmark collections. Although this study employed a reproducible benchmark consisting of a frozen metadata snapshot and fifteen representative search queries, larger query sets covering additional application domains and more diverse user information needs would provide a more comprehensive assessment of retrieval effectiveness. Future studies could also employ larger pools of relevance judgments, multiple independent assessors, and additional benchmark queries to further improve the robustness and external validity of the evaluation [39, 25, 16].

Another promising research direction is the automatic optimisation of the fuzzy inference system. The current implementation employs manually designed membership functions and expert-defined fuzzy rules to maintain transparency and interpretability. Future work could investigate adaptive optimisation techniques capable of learning membership functions, rule weights, or fuzzy rule bases directly from user interaction data while preserving explainability [42, 24, 38, 12].

The proposed framework could also be extended by integrating modern neural retrieval and learning-to-rank techniques. Recent advances in dense retrieval models, transformer-based reranking, and large language model ranking have demonstrated strong retrieval performance across numerous information retrieval tasks. Combining these approaches with transparent fuzzy reasoning may enable hybrid retrieval systems that balance retrieval effectiveness with interpretability, thereby supporting the objectives of Human-Centred Information Retrieval [33, 18, 19, 22, 37, 23].

Another important direction involves the development of personalised retrieval models. The current framework applies identical ranking criteria to all users regardless of their search objectives or domain expertise. Future systems could adapt fuzzy membership functions or ranking criteria according to user preferences, previous search behaviour, or contextual information while maintaining transparent explanations for ranking decisions. Such adaptive approaches would further strengthen the human-centred nature of the retrieval process [3, 26, 36].

Future work should also expand the formative user evaluation. The present study

provides an initial assessment involving ten Master’s students under a single evaluation condition. Larger studies involving participants from government agencies, research institutions, industry, and members of the public would provide stronger evidence regarding usability, transparency, explainability, and practical usefulness. Controlled experiments comparing the proposed HCFIR framework with conventional retrieval interfaces would further enable investigation of how explanation mechanisms influence user behaviour, interpretation of ranking decisions, and overall interaction with Open Government Data portals [5, 27, 36].

Finally, the methodology developed in this thesis could be validated across additional Open Government Data portals beyond Switzerland. Evaluating the proposed HCFIR framework using different metadata standards, portal architectures, languages, and organisational contexts would provide valuable evidence regarding its generalisability and practical applicability. Such studies would further support the development of transparent, explainable, and metadata-aware Human-Centred Information Retrieval systems for public-sector data discovery [15, 1, 6, 28].

8.9 Final Remarks

This thesis set out to investigate how HCIR can improve access to Swiss OGD through the integration of fuzzy logic-based multi-criteria ranking and explainable retrieval mechanisms. The research has demonstrated that effective IR is not solely determined by ranking accuracy, but also by the extent to which users can understand, interpret, and trust the retrieval process.

The proposed framework combines conventional information retrieval techniques with an interpretable Mamdani fuzzy inference system and metadata-driven explanation generation to support transparent and human-centred dataset discovery. Through its implementation and evaluation, this research demonstrates that fuzzy logic can effectively complement traditional keyword-based retrieval by incorporating multiple human-centred relevance criteria while maintaining competitive retrieval effectiveness.

Beyond the technical implementation, this research contributes to the broader objective of making Open Government Data more accessible, understandable, and usable. By providing interpretable explanations of the factors influencing ranking decisions, the proposed framework supports transparent exploration of public information and helps users interpret retrieved datasets more effectively.

Although opportunities remain for extending the framework through larger-scale evaluations, broader deployment, and integration with emerging retrieval technologies, the research establishes a practical foundation for future work on transparent and explainable HCIR.

In conclusion, this thesis demonstrates that Human-Centred Information Retrieval

can enhance Open Government Data discovery by integrating interpretable fuzzy multi-criteria ranking with transparent explanation mechanisms. Rather than replacing conventional lexical retrieval techniques, the proposed HCFIR framework complements existing retrieval infrastructures by supporting exploratory search and providing interpretable explanations for ranking decisions. Although the evaluation was conducted using a limited benchmark and a formative user evaluation with ten participants, the results indicate that combining technical retrieval evaluation with participant perceptions of transparency, explainability, and usability provides a more comprehensive assessment of retrieval quality than retrieval metrics alone. The proposed framework therefore establishes a practical, reproducible, and explainable foundation for future research on human-centred retrieval systems within Open Government Data environments.

Bibliography

- [1] Judie Attard, Fabrizio Orlandi, Simon Scerri, and Sören Auer. A systematic review of open government data initiatives. *Government Information Quarterly*, 32(4):399–418, 2015. doi: 10.1016/j.giq.2015.07.006.
- [2] Ricardo Baeza-Yates and Berthier Ribeiro-Neto. *Modern Information Retrieval*. 1999.
- [3] Nicholas J. Belkin. Human-centered information retrieval and interaction. *Information Research*, 20(3), 2015.
- [4] Gloria Bordogna and Gabriella Pasi. A fuzzy linguistic approach generalizing boolean information retrieval: A model and its evaluation. *Journal of the American Society for Information Science*, 44(2):70–82, 1993. doi: 10.1002/(SICI)1097-4571(199303)44:2<70::AID-ASI4>3.0.CO;2-H.
- [5] John Brooke. Sus: A "quick and dirty" usability scale. In *Usability Evaluation in Industry*, pages 189–194, 1996.
- [6] Adriane Chapman, Elena Simperl, Laura Koesten, George Konstantinidis, Luis-Daniel Ibáñez-Gonzalez, Emilia Kacprzak, and Paul Groth. Dataset search: A survey. *The VLDB Journal*, 29(1):251–272, 2020. doi: 10.1007/s00778-019-00564-x.
- [7] Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30, 2006. ISSN 1532-4435. URL <https://www.jmlr.org/papers/v7/demsar06a.html>.
- [8] Finale Doshi-Velez and Been Kim. Towards a rigorous science of interpretable machine learning. *arXiv preprint*, 2017.
- [9] Federal Statistical Office Switzerland. Open government data strategy switzerland, 2024. URL <https://www.bfs.admin.ch>.
- [10] Federal Statistical Office Switzerland. opendata.swiss: Swiss open government data portal, 2026. <https://opendata.swiss>.

- [11] Marti A. Hearst. *Search User Interfaces*. Cambridge University Press, New York, 2009. ISBN 9780521113793. doi: 10.1017/CBO9781139644082.
- [12] Enrique Herrera-Viedma. Modeling the retrieval process for an information retrieval system using an ordinal fuzzy linguistic approach. *Journal of the American Society for Information Science and Technology*, 52(6):460–475, 2001. doi: 10.1002/asi.1081.
- [13] Alan R. Hevner, Salvatore T. March, Jinsoo Park, and Sudha Ram. Design science in information systems research. *MIS Quarterly*, 28(1):75–105, 2004. doi: 10.2307/25148625.
- [14] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 1979.
- [15] Marijn Janssen, Yannis Charalabidis, and Anneke Zuiderwijk. Benefits, adoption barriers and myths of open data and open government. *Information Systems Management*, 29(4):258–268, 2012. doi: 10.2753/IMS0742-1222290402.
- [16] Kalervo Järvelin and Jaana Kekäläinen. Cumulated gain-based evaluation of ir techniques. *ACM Transactions on Information Systems*, 20(4):422–446, 2002. doi: 10.1145/582415.582418.
- [17] Thorhildur Jetzek, Michel Avital, and Niels Bjorn-Andersen. Generating value from open government data. *Information Polity*, 19(1-2):17–36, 2014. doi: 10.3233/IP-140328.
- [18] Vladimir Karpukhin et al. Dense passage retrieval for open-domain question answering. In *EMNLP*, 2020.
- [19] Omar Khattab and Matei Zaharia. Colbert: Efficient and effective passage search via contextualized late interaction over bert. In *SIGIR*, 2020.
- [20] Laura M. Koesten and Jaspreet Singh. Searching data portals – more complex than we thought? In *Proceedings of the CHIIR 2017 Workshop on Supporting Complex Search Tasks*, Oslo, Norway, 2017. URL <https://ceur-ws.org/Vol-1798/paper6.pdf>.
- [21] Donald H. Kraft and Cheryl Colvin. *Fuzzy Information Retrieval*. Springer, 2017. doi: 10.1007/978-3-662-05651-8.
- [22] Jimmy Lin, Rodrigo Nogueira, and Andrew Yates. *Pretrained Transformers for Text Ranking*. Morgan & Claypool, 2021.
- [23] Tie-Yan Liu. Learning to rank for information retrieval. *Foundations and Trends in Information Retrieval*, 3(3):225–331, 2009. doi: 10.1561/15000000016.

- [24] Ebrahim H. Mamdani and Sedrak Assilian. An experiment in linguistic synthesis with a fuzzy logic controller. *International Journal of Man-Machine Studies*, 7(1): 1–13, 1975. doi: 10.1016/S0020-7373(75)80002-2.
- [25] Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze. *Introduction to Information Retrieval*. Cambridge University Press, 2008.
- [26] Gary Marchionini. Exploratory search: From finding to understanding. *Communications of the ACM*, 49(4):41–46, 2006. doi: 10.1145/1121949.1121979.
- [27] Tim Miller. Explanation in artificial intelligence: Insights from the social sciences. *Artificial Intelligence*, 267:1–38, 2019. doi: 10.1016/j.artint.2018.07.007.
- [28] Sebastian Neumaier, Jürgen Umbrich, and Axel Polleres. Automated quality assessment of metadata across open data portals. *Journal of Data and Information Quality*, 8(1):1–29, 2016. doi: 10.1145/2964909.
- [29] Donald A. Norman. *The Design of Everyday Things*. Basic Books, New York, revised and expanded edition, 2013. ISBN 9780465050659.
- [30] Open Data Charter. International open data charter, 2015. URL <https://opendatacharter.net>.
- [31] Open Knowledge Foundation. Open definition 2.1, 2024. URL <https://opendefinition.org>.
- [32] Ken Peffers, Tuure Tuunanen, Marcus A. Rothenberger, and Samir Chatterjee. A design science research methodology for information systems research. *Journal of Management Information Systems*, 24(3):45–77, 2007. doi: 10.2753/MIS0742-1222240302.
- [33] Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *EMNLP*, pages 3982–3992, 2019. doi: 10.18653/v1/D19-1410.
- [34] Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: Bm25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009. doi: 10.1561/15000000019.
- [35] Gerard Salton and Christopher Buckley. Term-weighting approaches in automatic text retrieval. *Information Processing and Management*, 24(5):513–523, 1988. doi: 10.1016/0306-4573(88)90021-0.

- [36] Ben Shneiderman. Human-centered ai and human-centered information retrieval. *International Journal of Human-Computer Interaction*, 38(6):494–507, 2022. doi: 10.1080/10447318.2021.1995443.
- [37] Wei Sun et al. Is chatgpt good at search? investigating large language models as re-ranking agents. In *EMNLP*, 2023.
- [38] T. Takagi and M. Sugeno. Fuzzy identification of systems and its applications to modeling and control. *IEEE Transactions on Systems, Man, and Cybernetics*, 15(1): 116–132, 1985. doi: 10.1109/TSMC.1985.6313399.
- [39] Ellen M. Voorhees. The trec-8 question answering track report. 1999.
- [40] Ryen W. White and Resa A. Roth. Exploratory search: Beyond the query-response paradigm. *Synthesis Lectures on Information Concepts, Retrieval, and Services*, 1(1):1–98, 2009. doi: 10.2200/S00174ED1V01Y200901ICR003.
- [41] Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1945.
- [42] Lotfi A. Zadeh. Fuzzy sets. *Information and Control*, 8(3):338–353, 1965. doi: 10.1016/S0019-9958(65)90241-X.
- [43] Hans-Jürgen Zimmermann. *Fuzzy Set Theory and Its Applications*. Springer, 4 edition, 2010. doi: 10.1007/978-94-010-0646-0.

Appendix A

Evaluation Questionnaire

This appendix reproduces the questionnaire administered during the user study. The questionnaire consisted of participant background questions, three retrieval tasks, twenty-one five-point Likert-scale evaluation items covering usability, retrieval effectiveness, explainability, trust, and overall satisfaction, followed by four open-ended feedback questions. Participants rated each evaluation statement using a five-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree), with an additional “N/A” option available when a question was not applicable.

Participant Information

Participants first completed a short demographic questionnaire consisting of the following items:

1. Email address.
2. Age.
3. Current background (Bachelor’s/Master’s student, PhD student, Researcher, Other).
4. Familiarity with Open Government Data portals.
5. Frequency of searching for datasets or Open Government Data.

Evaluation Tasks

Before completing the evaluation questionnaire, participants performed the following dataset retrieval tasks using the developed HCIR prototype.

1. **Structured Search Task**

Search for a government dataset related to transport or mobility and identify one dataset that would be useful for analysis.

2. Exploratory Search Task

Perform a broad or vague search (e.g., environment, population, climate, or transport) and explore whether the system helps identify relevant datasets.

3. Free Exploration Task

Freely explore the search system for approximately 10–15 minutes using topics of personal interest and pay particular attention to the relevance of search results, usefulness of explanations, ease of use, clarity of ranking information, and overall user experience.

A. Usability

1. The interface is easy to understand.
2. It was easy to enter and modify search queries.
3. The search process felt intuitive.
4. The layout of the search results was clear.
5. The system provided helpful feedback when searching or refining a query.

B. Retrieval Effectiveness

6. The result cards provided enough information to decide whether a dataset was relevant.
7. The displayed metadata helped me judge the usefulness of a dataset.
8. It was easy to refine or modify my search query.
9. The system helped me recover when my first search query did not return good results.
10. The returned datasets were relevant to my search queries.
11. The most relevant datasets usually appeared near the top of the results.
12. The system handled broad or exploratory queries effectively.
13. The system helped me find useful results even when my query was vague, incomplete, or imprecise.
14. I could easily identify which dataset was most relevant to my needs.

C. Explainability

15. The explanations for ranking decisions were understandable.
16. The relevance scores helped me understand why certain datasets were ranked highly.
17. The explanations made the ranking feel more transparent.
18. The displayed metadata, such as tags, organization, and update date, helped me judge dataset relevance.
19. The system provided enough information for me to trust the search results.

D. Overall Satisfaction

20. Overall, I am satisfied with the system.
21. I would use a system like this for finding government datasets.

Open Feedback Questions

Participants were also invited to provide qualitative feedback through the following open-ended questions.

1. What did you like most about the system?
2. What aspects were confusing or difficult to use?
3. What improvements would you suggest?
4. Any additional comments?

Appendix B

Example Fuzzy Rules

The following examples illustrate the fuzzy inference rules used within the proposed retrieval framework.

1. IF BM25 Similarity is High AND Metadata Completeness is High AND Dataset Recency is High AND Resource Availability is High THEN Relevance is Very High.
2. IF BM25 Similarity is High AND Metadata Completeness is Medium THEN Relevance is High.
3. IF BM25 Similarity is Medium AND Metadata Completeness is High THEN Relevance is High.
4. IF BM25 Similarity is Medium AND Dataset Recency is High THEN Relevance is Medium.
5. IF BM25 Similarity is Low THEN Relevance is Low.

B.1 Calibrated Membership Function Parameters

The Mamdani Fuzzy Inference System employs data-calibrated membership functions derived from the statistical characteristics of the evaluation corpus. Rather than selecting linguistic boundaries manually, the calibration process determines breakpoints from the observed distributions of metadata completeness, dataset recency, and resource availability. Similarity and the output relevance variable use predefined membership functions to preserve a stable interpretation of textual relevance across retrieval tasks. Table [B.1](#) summarises the calibrated membership functions used by the prototype.

Table B.1: Membership functions used by the Mamdani Fuzzy Inference System.

Variable	Linguistic Term	Type	Parameters
Similarity	Not Relevant	Triangular	(0.000, 0.000, 0.150)
	Somewhat Relevant	Triangular	(0.100, 0.250, 0.400)
	Relevant	Triangular	(0.300, 0.500, 0.700)
	Highly Relevant	Triangular	(0.600, 0.780, 0.920)
	Exact Match	Triangular	(0.850, 0.950, 1.000)
Metadata Completeness	Low	Triangular	(0.000, 0.000, 0.564)
	Partial	Triangular	(0.564, 0.649, 0.814)
	Medium	Trapezoidal	(0.649, 0.814, 0.819, 0.865)
	High	Trapezoidal	(0.814, 0.819, 0.865, 1.000)
	Complete	Triangular	(0.865, 1.000, 1.000)
Dataset Recency	Very Recent	Triangular	(1.000, 1.000, 1.000)
	Recent	Trapezoidal	(1.000, 1.000, 1.000, 49.500)
	Moderate	Trapezoidal	(1.000, 49.500, 98.000, 98.000)
	Old	Trapezoidal	(49.500, 98.000, 98.000, 98.000)
	Very Old	Triangular	(98.000, 98.000, 98.000)
Resource Availability	Minimal	Triangular	(0.000, 0.000, 1.000)
	Limited	Triangular	(0.000, 1.000, 2.000)
	Moderate	Triangular	(1.000, 2.000, 2.000)
	Rich	Trapezoidal	(2.000, 2.000, 4.000, 4.550)
	Comprehensive	Triangular	(4.000, 4.550, 6.000)
Overall Relevance	Very Low	Triangular	(0.000, 0.000, 0.250)
	Low	Triangular	(0.100, 0.280, 0.450)
	Moderate	Triangular	(0.350, 0.500, 0.650)
	Good	Triangular	(0.550, 0.720, 0.880)
	Excellent	Triangular	(0.800, 0.920, 1.000)

Appendix C

Example Explanation Trace

This appendix presents one complete explanation trace generated by the final production implementation of the proposed HCIR framework. The example is taken directly from the system output without manual modification and illustrates how the fuzzy inference engine produces an explainable ranking decision for a retrieved dataset.

C.1 Query

- Query ID: M0B-04
- Query: *Veloverkehr und Radwege in der Schweiz*

C.2 Retrieved Dataset

- Dataset ID: 3f8295e7-6758-44a9-b1b5-6cabfa5241e5
- Dataset Title (German): *Bildersammlung Max van Berchem*
- Dataset Title (English): *Max van Berchem image collection*

C.3 Input Values Used by the Fuzzy Inference System

Input Variable	Value
Thematic Similarity	0.1815
Recency	0.0000
Metadata Completeness	0.7500
Resource Availability	3.0000

Table C.1: Input values supplied to the fuzzy inference engine.

C.4 Fuzzification

The continuous input values were converted into linguistic membership degrees.

Variable	Linguistic Term	Membership Degree
Thematic Similarity	Somewhat Relevant	0.2099
Thematic Similarity	Not Relevant	0.0926
Recency	Very Recent	1.0000
Metadata Completeness	Mostly Complete	0.3333
Resource Availability	Limited	0.6667

Table C.2: Non-zero membership degrees generated during fuzzification.

C.5 Activated Fuzzy Rules

Only two rules contributed to the final decision.

Rule	Description	Strength
16	IF thematic similarity is somewhat relevant THEN relevance is low	0.2099
14	IF thematic similarity is not relevant THEN relevance is very low	0.0926

Table C.3: Dominant fuzzy rules activated for the selected dataset.

The remaining rules had zero firing strength and therefore did not influence the final decision.

C.6 Defuzzification

The aggregated fuzzy output was converted into a crisp relevance score using the implemented defuzzification procedure.

- Crisp Output: 22.6719
- Final Relevance Score Presented to the User: 22.7/100

C.7 Explanation Presented to the User

The prototype displayed the following explanation:

Final Relevance Score: 22.7/100

Input Analysis

Thematic similarity: 0.18 $\rightarrow$ “somewhat relevant” ($\mu = 0.21$)

Recency: 0.00 $\rightarrow$ “very recent” ($\mu = 1.00$)

Metadata completeness: 0.75 $\rightarrow$ “mostly complete” ($\mu = 0.33$)

Resource availability: 3.00 $\rightarrow$ “limited” ($\mu = 0.67$)

Top Contributing Rules

1. IF thematic similarity IS somewhat relevant THEN relevance score IS low

Strength: 0.21

Reason: Weak thematic matches should still be handled by fuzzy rules.

2. IF thematic similarity IS not relevant THEN relevance score IS very low

Strength: 0.09

Reason: Non-relevant datasets always rank very low.

C.8 Traceability

This appendix is derived directly from the production execution trace generated by the final implementation. The trace records the numerical fuzzy inputs, membership degrees, rule activations, dominant rules, crisp output, and explanation presented to the user. The current implementation does not expose the individual BM25 matched query terms or matched metadata fields in the explanation trace; therefore, these values are not reported here.